# PROJECT REPORT

## Service Center Management System

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## T.N. RAO COLLEGE, RAJKOT

Master of Science in Information Technology and Computer Applications

**Branch:** Computer Science

**Year:** 2025-26

# T. N. Rao College of I.T. - Rajkot

**B/h. MCA Department, Nr. Saurashtra University Campus,**
**Rajkot - 360 005, Ph.0281-2970471, Fax : 0281-2970472**


## CERTIFICATE

This is to certify that Mr/Miss **Het Agravat** of M.Sc. (IT & CA) has successfully completed his / her 1st Semester Project work on **SCMS** During the academic year 2025-26.

Project Guide

Asif Yusufzai
HOD

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# Abstract

The **Service Center Management System (SCMS)** is a **comprehensive web-based solution** developed to automate and optimize the operations of service centers dealing with **home appliance repairs and maintenance**. By centralizing the processes of complaint registration, technician assignment, and status tracking, SCMS significantly improves operational efficiency, reduces delays, and enhances overall customer satisfaction.

## Key Objectives:

- **Streamline Operations:** The primary goal of SCMS is to automate the manual processes of service centers, including complaint management, technician assignments.
- **Improve Efficiency:** By reducing the time taken to assign technicians and track complaints, the system helps service centers respond to customer requests faster.
- **Enhance Transparency:** Both Admins and Dealers have clear, real-time visibility into complaint statuses, technician assignments, and pending tasks.

## System Users:

1. **Admin:**
    - **Full Control:** Admins have complete control over the system, including managing complaints, technicians, dealers, and customer information.
    - **Technician Management:** Admins can assign complaints to technicians, track their progress, and update complaint statuses.
    - **Dealer Management:** Admins can add and manage dealers, including providing them with credentials and controlling their access.
    - **Complaint Management:** Admins can view, update, assign, and close complaints based on technician feedback.

2. **Dealer:**
    - **Complaint Registration:** Dealers can easily register new complaints for customers using a simplified, user-friendly interface.
    - **Complaint Tracking:** Dealers have access to the status of complaints they've registered, including updates on technician assignments and complaint resolution.
    - **View Complaint History:** Dealers can track all previous complaints and view their status histories.

## Features and Functionalities:

- **Complaint Management:** The system provides tools to register, assign, update, and track complaints for a range of home appliances, such as air conditioners, refrigerators, and washing machines.
- **Technician Assignment:** Admins can efficiently assign technicians to customer complaints based on availability and expertise, ensuring prompt service.
- **Real-time Tracking:** Both Admins and Dealers can monitor the progress of complaints, with status updates being provided throughout the lifecycle of the complaint.
- **Technician and Dealer Database:** SCMS maintains a comprehensive database of technicians and dealers, including contact information, performance tracking, and history of assigned tasks.
- **User-Friendly Interface:** Built with Bootstrap for responsive design, the system ensures a smooth user experience on both desktop and mobile devices.

## Benefits:

- **Time-Saving:** Automating complaint handling and technician assignment saves valuable time, allowing service centers to focus on delivering quality service.
- **Cost-Effective:** The system reduces the need for manual processes and paperwork, cutting down on operational costs.
- **Scalable:** Designed to handle growing data volumes, SCMS can scale up to accommodate an increasing number of complaints, technicians, and dealers without compromising performance.
- **Customizable Dashboards:** Both Admins and Dealers have tailored dashboards that provide relevant information, such as complaint statistics, technician availability, and dealer activity.

In summary, SCMS represents a robust and scalable solution that enhances the efficiency of home appliance service centers, ensuring faster service delivery, higher customer satisfaction, and streamlined operations.

# Preface

This document serves as a comprehensive guide to the development and implementation of the **Service Center Management System (SCMS)**. The project was initiated with the aim of **enhancing the efficiency and accuracy** of complaint management in service centers dealing with home appliances such as air conditioners, refrigerators, and washing machines.

## Purpose of the Document:

- To document the entire development process of the SCMS, providing an in-depth overview of the technologies, methodologies, and tools used.
- To serve as a reference for future enhancements and modifications to the system.
- To outline the challenges faced during development and how they were addressed.
- To showcase the system's architecture, design, and implementation to facilitate understanding for developers, project managers, and stakeholders.

## Development Approach:

The SCMS project was developed using a **core PHP-based architecture** with **MySQL** as the backend database. The front end was built using **HTML**, **CSS**, **JavaScript**, and **Bootstrap**, ensuring a responsive and user-friendly interface. Throughout the development process, we adhered to the **Software Development Life Cycle (SDLC)**, ensuring that all phases—**requirement analysis, system design, development, testing, and deployment**—were properly documented and executed.

## Project Goals:

- **Automate Complaint Management:** To replace manual processes with a web-based system for better management of customer complaints and technician assignments.
- **Centralize Data:** To ensure that all relevant data—complaints, technician assignments, and dealer records—are centrally stored and accessible to authorized users.
- **Optimize Service Delivery:** By reducing delays in technician assignment and complaint resolution, the system aims to increase overall customer satisfaction.
- **Ensure Scalability:** SCMS was designed to handle growing numbers of complaints and users without a drop in performance, ensuring its adaptability to larger-scale implementations.

## Project Methodology:

The **Software Development Life Cycle (SDLC)** model was rigorously followed to ensure that the system met the project requirements effectively and efficiently. The key phases of SDLC, including **planning, requirement analysis, system design, implementation, testing, and maintenance**, were carried out methodically. Each phase included detailed documentation, ensuring transparency and traceability throughout the development process.

# Project Introduction

The **Service Center Management System (SCMS)** is a web-based application developed to address the challenges faced by home appliance service centers in managing customer complaints, assigning technicians, and overseeing dealer operations. The system automates the entire service process, from complaint registration to resolution, replacing manual, error-prone methods with a more efficient and streamlined solution.

## Problem Statement:

In traditional service centers, managing customer complaints, tracking technician assignments, and keeping dealers informed is often a manual process, leading to delays, errors, and overall inefficiency. These issues result in poor customer satisfaction and a lack of transparency in the service delivery process. SCMS was developed to overcome these challenges by offering a centralized and automated system that integrates all aspects of service center operations.

## Project Objectives:

- **Automate Complaint Management:** Simplify the process of registering, assigning, and tracking complaints to minimize delays and errors.
- **Centralize Data:** Store all relevant data—such as complaint details, technician assignments, and dealer activities—in a single system for easier access and management.
- **Optimize Technician Assignment:** Allow Admins to efficiently assign the most appropriate technicians based on availability and expertise.
- **Enhance Customer Satisfaction:** By streamlining the service process, the system ensures faster resolutions, which leads to improved customer experience.
- **Support Scalability:** Ensure the system is scalable to handle an increasing number of complaints, users, and dealers as the service center grows.

## System Overview:

The SCMS is designed to provide separate interfaces and functionalities for two types of users: **Admins** and **Dealers**.

## Key Features:

1. **Admin Panel:**
    - The Admin Panel is the core management hub of the system. Admins have access to various modules that allow them to oversee complaints, dealers, and technicians.
    - **Complaint Management:** Admins can register new complaints, assign technicians, update complaint statuses (e.g., new, assigned, closed), and monitor complaint history.
    - **Technician Management:** Admins can add and update technicians, track their availability, and assign them to relevant complaints based on their expertise and workload.
    - **Dealer Management:** Admins can register new dealers, update their information, and provide them with login credentials to access the system.
2. **Dealer Panel:**
    - The Dealer Panel provides dealers with a simplified interface for handling customer complaints.
    - **Complaint Registration:** Dealers can quickly register new complaints on behalf of their customers, ensuring timely response to service issues.
    - **Complaint Status Tracking:** Dealers can view the status of their registered complaints and track the resolution process in real time.
    - **Complaint History:** Dealers have access to a full history of their past complaints, providing transparency and allowing them to keep customers informed.

3. **Technician Assignment:**
    - The system enables Admins to assign technicians to specific complaints based on their availability and skill set.
    - Technicians' performance and status can be monitored, ensuring effective management of resources.
4. **Complaint Tracking:**
    - The system provides real-time tracking of complaint statuses for both Admins and Dealers. Updates on each stage of the complaint lifecycle—from registration to resolution—are clearly displayed, ensuring transparency in the process.
5. **Scalability:**
    - SCMS is designed with scalability in mind. As the service center grows, the system can easily accommodate a larger volume of complaints, technicians, and dealers without compromising performance.
    - This flexibility allows the system to be adaptable to different sizes and scales of service centers.

## Conclusion:

The Service Center Management System (SCMS) addresses the operational inefficiencies in traditional service centers by offering an automated, organized, and centralized solution for managing customer complaints, technician assignments, and dealer operations. By implementing SCMS, service centers can significantly improve their operational efficiency, reduce delays, and enhance customer satisfaction, while also ensuring that the system can grow alongside their business.

# Software and Hardware Requirements

## Software Requirements

To ensure optimal performance and compatibility, the following software requirements are necessary for the development, testing, and deployment of the **Service Center Management System (SCMS)**:

1. **Operating System:**
    - The system is compatible with multiple platforms, including:
        - **Windows** (7 or higher)
        - **Linux** (any modern distribution such as Ubuntu, CentOS)
        - **MacOS** (10.12 or higher)
    - These operating systems provide the flexibility for developers and users to work in their preferred environments.

2. **Web Server:**
    - The SCMS requires a web server for hosting and running the application. Recommended web server options include:
        - **Apache HTTP Server**
        - This can be easily set up using:
            - **XAMPP** (cross-platform solution for Windows, Linux, MacOS)
            - **WAMP** (Windows-based solution)
    - Apache serves as the backbone for hosting the PHP scripts and managing incoming HTTP requests.

3. **Database:**
    - **MySQL** is used as the relational database for storing all relevant data related to complaints, dealers, technicians, and users.
    - MySQL provides the necessary performance and scalability to handle both small and large-scale data operations efficiently.

4. **Server-Side Language:**
    - **PHP** (version 7.0 or higher) is the primary server-side scripting language used to develop and run the SCMS.
    - PHP handles core business logic, interacts with the database, and processes requests from users via the web server.

5. **Client-Side Technologies:**
    - The user interface of the SCMS is built using the following technologies:
        - **HTML**: Provides the structure and layout of the web pages.
        - **CSS**: Ensures a responsive and visually appealing design for the user interface.
        - **JavaScript (with jQuery)**: Adds dynamic functionality and enhances user interactions, such as form validation and real-time updates without refreshing the page.

6. **Frameworks/Libraries:**
    - **Bootstrap**: Utilized for creating a responsive and mobile-friendly user interface. Bootstrap's grid system and ready-to-use components make it easy to design forms, buttons, tables, and other UI elements.
    - **DataTables**: A jQuery plugin that enhances HTML tables by providing features such as search, pagination, sorting, and filtering for better data handling within the SCMS.

## Hardware Requirements

The following hardware specifications are recommended to ensure smooth performance and reliability of the SCMS:

1. **Processor:**
    - A modern processor with sufficient computational power is required to run both the web server and database efficiently. The minimum requirement is:
        - **Intel Core i3 or higher** (or equivalent AMD processor)
        - For larger deployments or high-traffic systems, upgrading to an **Intel Core i5/i7** or higher is recommended to handle increased workloads.
2. **RAM:**
    - To ensure fast data processing and optimal application performance, the system requires:
        - **4 GB or more** of RAM for basic usage.
        - **8 GB or more** is recommended for handling larger datasets or multiple concurrent users.
3. **Storage:**
    - Adequate storage space is necessary for storing system files, databases, and server-side scripts. The recommended free storage is:
        - **100 GB** free space.
        - This allows room for database growth and any additional updates or logs generated by the system over time.
4. **Internet Connectivity:**
    - While the system can run on a local server, **internet access** is necessary for:
        - **Deployment**: Hosting the system on a web server for access by remote users (Admins, Dealers).
        - **Updates**: Downloading software updates, patches, and new features.

# Tools and Technology Used

The **Service Center Management System (SCMS)** project was developed using a range of tools and technologies, carefully chosen to ensure performance, scalability, and ease of development.

## Programming Language:

- **PHP (Core)**:
    - The backend logic of the SCMS is entirely developed using **Core PHP**. PHP is a widely-used, open-source scripting language that is especially suited for web development.
    - Core PHP was selected to ensure that the project remains lightweight, modular, and easily maintainable without relying on heavy frameworks.

## Database:

- **MySQL**:
    - **MySQL** is used as the relational database to manage and store data such as user accounts, complaints, technicians, and dealers.
    - It provides efficient querying capabilities and supports all operations required by the SCMS, such as storing, updating, and retrieving complaint data and system-related information.

## Frontend Technologies:

- **HTML (Hypertext Markup Language)**:
    - HTML is used to structure the web pages and forms, ensuring that the user interface is well-organized and accessible across different devices.
- **CSS (Cascading Style Sheets)**:
    - CSS is used for styling the SCMS application, ensuring a consistent look and feel across all pages. It allows customization of layouts, colors, and fonts to create a professional user experience.
- **JavaScript (with AJAX)**:
    - JavaScript powers dynamic elements on the client side. It handles form validation, page interactions, and other client-side logic.
    - **AJAX (Asynchronous JavaScript and XML)** is used for handling background requests without reloading the page. This enables smooth updates, especially for real-time complaint management and data fetching.

- **Bootstrap**:
    - **Bootstrap** is a front-end framework used for creating responsive, mobile-first web designs. It simplifies the process of building forms, tables, buttons, and other UI components with its pre-built CSS and JavaScript components.
    - This ensures that the SCMS is compatible across multiple devices and screen sizes.

## Backend Technologies:

- **PHP**:
    - PHP is not only used for the backend logic but also for interacting with the MySQL database, handling CRUD operations (Create, Read, Update, Delete), and processing HTTP requests.
- **MySQL**:
    - MySQL serves as the database where all data related to users, complaints, technicians, and dealers is stored. The backend PHP code interacts with the MySQL database through queries for fetching, inserting, and updating information.

## Development Tools:

- **Sublime Text**:
    - A lightweight text editor used during development for writing and editing code. It provides useful features such as syntax highlighting, auto-completion, and project-wide search, enhancing productivity.
- **Visual Studio Code**:
    - Another powerful code editor, **Visual Studio Code** was used in this project for more advanced features like integrated debugging, extensions for PHP, JavaScript, and Git support. It also facilitates project collaboration through Git integration.
- **MySQL Workbench**:
    - **MySQL Workbench** is used for database design, modeling, and management. It provides a user-friendly interface for creating and managing database schemas, running queries, and optimizing database performance.

## Version Control:

- **Git**:
    - Git is the version control system used for tracking changes during the development of the SCMS. It helps in maintaining different versions of the project, facilitates collaboration among developers, and allows easy rollback to previous versions when needed.

## Libraries:

- **DataTables**:
    - **DataTables** is a jQuery plugin used for enhancing HTML tables with features like pagination, search, sorting, and filtering. It makes managing large datasets (such as complaints, technicians, and dealers) much easier for users, ensuring fast access and display of data.
- **jQuery**:
    - **jQuery** is a JavaScript library that simplifies DOM manipulation, event handling, and AJAX requests. It is used extensively in the SCMS to handle client-side tasks, ensuring smooth interactions and seamless communication with the backend server.

# System Analysis and Design

This section details the architecture, data flow, and design of the Service Center Management System (SCMS). It highlights how the system processes data and how the various entities within the system interact.

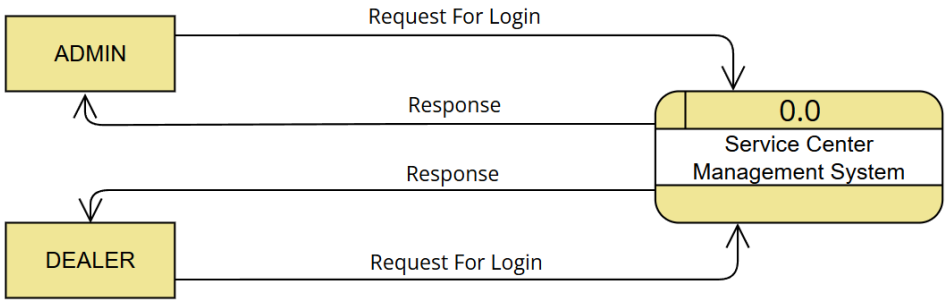
## Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) provides a graphical representation of the data movement within the SCMS. It illustrates how the different modules of the system interact and exchange data.

### Level 0 DFD:

The **Level 0 DFD** represents the **high-level view** of the system, showcasing the primary interactions between the key entities: Admin, Dealer, and the SCMS.

**Context Level DFD (0 Level)**



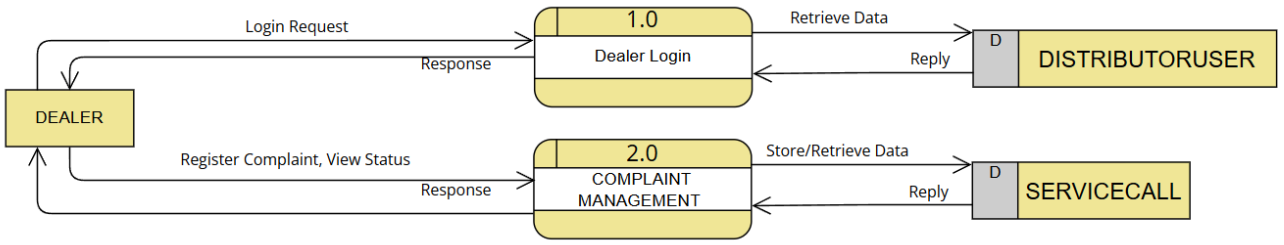

0 Level DFD For Service Center Management System

- **Admin**: Manages complaints, dealers, and technicians.
    - Interacts with the system to register and assign complaints to technicians.
    - Updates complaint statuses such as "Assigned," "Closed," or "Cancelled."
- **Dealer**: Registers new complaints and views the status of existing complaints.
    - Interacts with the system to track the complaint status and history.
- **System**: Central hub where all data is processed and stored. The system handles operations related to complaints, technician assignments, and dealer management.

## Level 1 DFD:

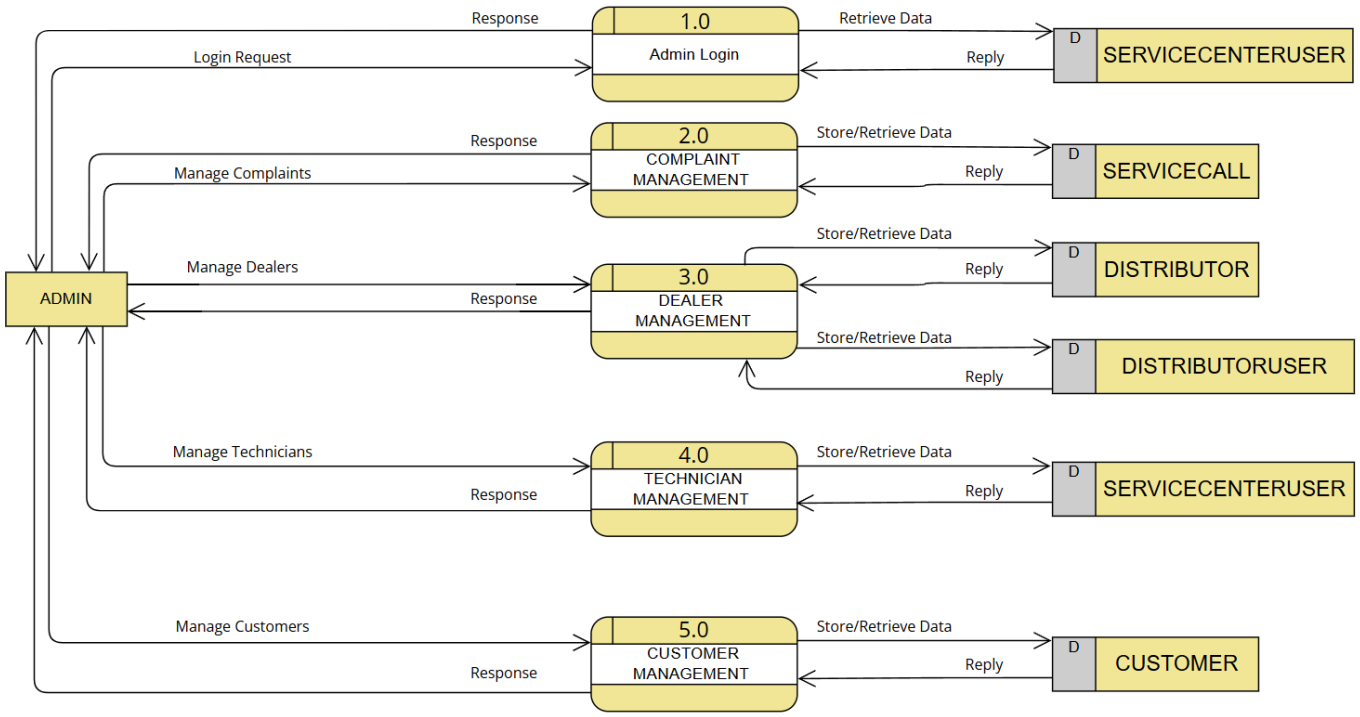
The **Level 1 DFD** breaks down the system's operations into its core modules and illustrates the **flow of data** between them.

**1st Level Dealer Side DFD**




1 Level DFD For SCMS Dealer

**1st Level Admin Side DFD**




1 Level DFD For SCMS Admin

## Data Dictionary

The Data Dictionary outlines the structure and design of the database used by the SCMS. It describes the key **tables, fields, and relationships** that are essential to the system's functioning.

| servicecenter.customer | | | |
|---|---|---|---|
| **Field Name** | **Data Type** | **Constraints** | **Description** |
| id | INT | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for the customer |
| CustomerName | VARCHAR(100) | NOT NULL | Name of the customer |
| CustomerAddress | VARCHAR(100) | NOT NULL | Address of the customer |
| CustomerCity | VARCHAR(45) | NOT NULL | City of the customer |
| MobileNumber | VARCHAR(45) | NOT NULL | Primary mobile number of the customer |
| WhatsAppNumber | VARCHAR(45) | NULL | WhatsApp number of the customer |
| ProductCompanyName | VARCHAR(45) | NULL | Name of the company that manufactured the product |
| ProductNameOrModel | VARCHAR(45) | NULL | Name or model of the product |
| LastMessageSendDate | DATE | NULL | Date of the last message sent to the customer |
| CreateDate | DATE | NOT NULL | Date when the customer record was created |
| CreateBy | INT | NOT NULL (References servicecenteruser.id) | ID of the user who created the customer record |
| ModifiedDate | DATE | NULL | Date when the customer record was last modified |
| ModifiedBy | INT | NOT NULL (References servicecenteruser.id) | ID of the user who last modified the customer record |
| IsActive | CHAR(1) | NOT NULL | Indicates whether the customer is active or inactive |

| servicecenter.servicecenteruser | | | |
|---|---|---|---|
| **Field Name** | **Data Type** | **Constraints** | **Description** |
| id | INT | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for the service center user |
| username | VARCHAR(45) | NOT NULL, UNIQUE | Username of the service center user |
| password | VARCHAR(100) | NOT NULL | Password of the service center user |
| primarymobileno | VARCHAR(10) | NOT NULL | Primary mobile number of the service center user |
| secondmobileno | VARCHAR(10) | NULL | Secondary mobile number of the service center user |
| firstname | VARCHAR(20) | NOT NULL | First name of the service center user |
| lastname | VARCHAR(20) | NULL | Last name of the service center user |
| address | VARCHAR(100) | NULL | Address of the service center user |
| city | VARCHAR(45) | NOT NULL | City of the service center user |
| roletype | VARCHAR(20) | NOT NULL | Role of the service center user (e.g., admin, technician) |
| isactive | CHAR(1) | NOT NULL | Indicates whether the service center user is active or inactive |
| createdate | DATETIME | NOT NULL | Date and time when the service center user was created |
| modifydate | DATETIME | NULL | Date and time when the service center user was last modified |

| servicecenter.distributor | | | |
|---|---|---|---|
| **Field Name** | **Data Type** | **Constraints** | **Description** |
| id | INT | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for the distributor |
| name | VARCHAR(100) | NOT NULL, UNIQUE | Name of the distributor |
| isactive | CHAR(1) | NOT NULL | Indicates whether the distributor is active or inactive |
| primarymobileno | VARCHAR(10) | NOT NULL | Primary mobile number of the distributor |
| secondmobileno | VARCHAR(10) | NULL | Secondary mobile number of the distributor |
| emailaddress | VARCHAR(100) | NULL | Email address of the distributor |
| address | VARCHAR(100) | NULL | Address of the distributor |
| city | VARCHAR(45) | NOT NULL | City of the distributor |

| servicecenter.distributoruser | | | |
|---|---|---|---|
| **Field Name** | **Data Type** | **Constraints** | **Description** |
| id | INT | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for the distributor user |
| username | VARCHAR(45) | NOT NULL, UNIQUE | Username of the distributor user |
| userpassword | VARCHAR(45) | NOT NULL | Password of the distributor user |
| isactive | CHAR(1) | NOT NULL | Indicates whether the distributor user is active or inactive |
| distributorid | INT | NOT NULL, FOREIGN KEY (References distributor.id) | ID of the distributor associated with the user |
| firstname | VARCHAR(45) | NULL | First name of the distributor user |
| lastname | VARCHAR(45) | NULL | Last name of the distributor user |
| mobileno | VARCHAR(25) | NULL | Mobile number of the distributor user |
| roletype | VARCHAR(10) | NOT NULL | Role of the distributor user (e.g., admin, technician) |
| createdate | DATETIME | NOT NULL | Date and time when the distributor user was created |
| modifydate | DATETIME | NULL | Date and time when the distributor user was last modified |

| servicecenter.servicecall | | | |
|---|---|---|---|
| **Field Name** | **Data Type** | **Constraints** | **Description** |
| id | INT | PRIMARY KEY, AUTO_INCREMENT | Unique identifier for the service call |
| callnumber | VARCHAR(45) | NOT NULL, UNIQUE | Call number of the service call |
| customername | VARCHAR(100) | NOT NULL | Name of the customer |
| customermobileno | VARCHAR(45) | NOT NULL | Mobile number of the customer |
| customeraddress | VARCHAR(100) | NOT NULL | Address of the customer |
| customercity | VARCHAR(45) | NOT NULL | City of the customer |
| calltype | VARCHAR(45) | NOT NULL | Type of the service call (e.g., repair, installation) |
| serviceworktype | VARCHAR(45) | NULL | Type of service work performed |
| paymenttype | VARCHAR(45) | NULL | Payment type used for the service call |
| calldate | DATETIME | NOT NULL | Date and time when the service call was created |
| callassigndate | DATETIME | NULL | Date and time when the service call was assigned to a technician |
| technicianassigned | INT | NULL (References servicecenteruser.id) | ID of the technician assigned to the service call |
| callcompletedate | DATETIME | NULL | Date and time when the service call was completed |
| callstatus | VARCHAR(45) | NOT NULL | Status of the service call (e.g., new, assigned, closed, canceled) |
| totalamount | DECIMAL(10,2) | NULL | Total amount of the service call |
| discountamount | DECIMAL(10,2) | NULL | Discount amount applied to the service call |
| finalamount | DECIMAL(10,2) | NULL | Final amount after applying discount |
| createdate | DATETIME | NOT NULL | Date and time when the service call was created |
| createby | INT | NULL (References servicecenteruser.id) | ID of the user who created the service call |
| createby_distributor_id | INT | NULL (References distributor.id) | ID of the distributor who created the service call |
| createby_distributoruser_id | INT | NULL (References distributoruser.id) | ID of the distributor user who created the service call |
| modifiedby | INT | NULL (References servicecenteruser.id) | ID of the user who last modified the service call |
| modifieddate | DATETIME | NULL | Date and time when the service call was last modified |
| customerproblem | VARCHAR(200) | NULL | Description of the customer's problem |
| callresolution | VARCHAR(500) | NULL | Resolution provided for the service call |

# Project Implementation

The Service Center Management System (SCMS) was implemented using a combination of **core PHP** and **MySQL** for the backend, ensuring a robust and scalable data handling mechanism. The frontend was developed using **HTML**, **CSS**, **JavaScript**, and **Bootstrap**, providing a responsive and user-friendly interface for both Admin and Dealer users.

## Backend Implementation:

- **Core PHP**: PHP was used to handle server-side logic, including data processing, validation, and communication with the database.
- **MySQL**: MySQL handled the storage and retrieval of data, allowing for efficient database management for complaints, technicians, dealers, and admins.

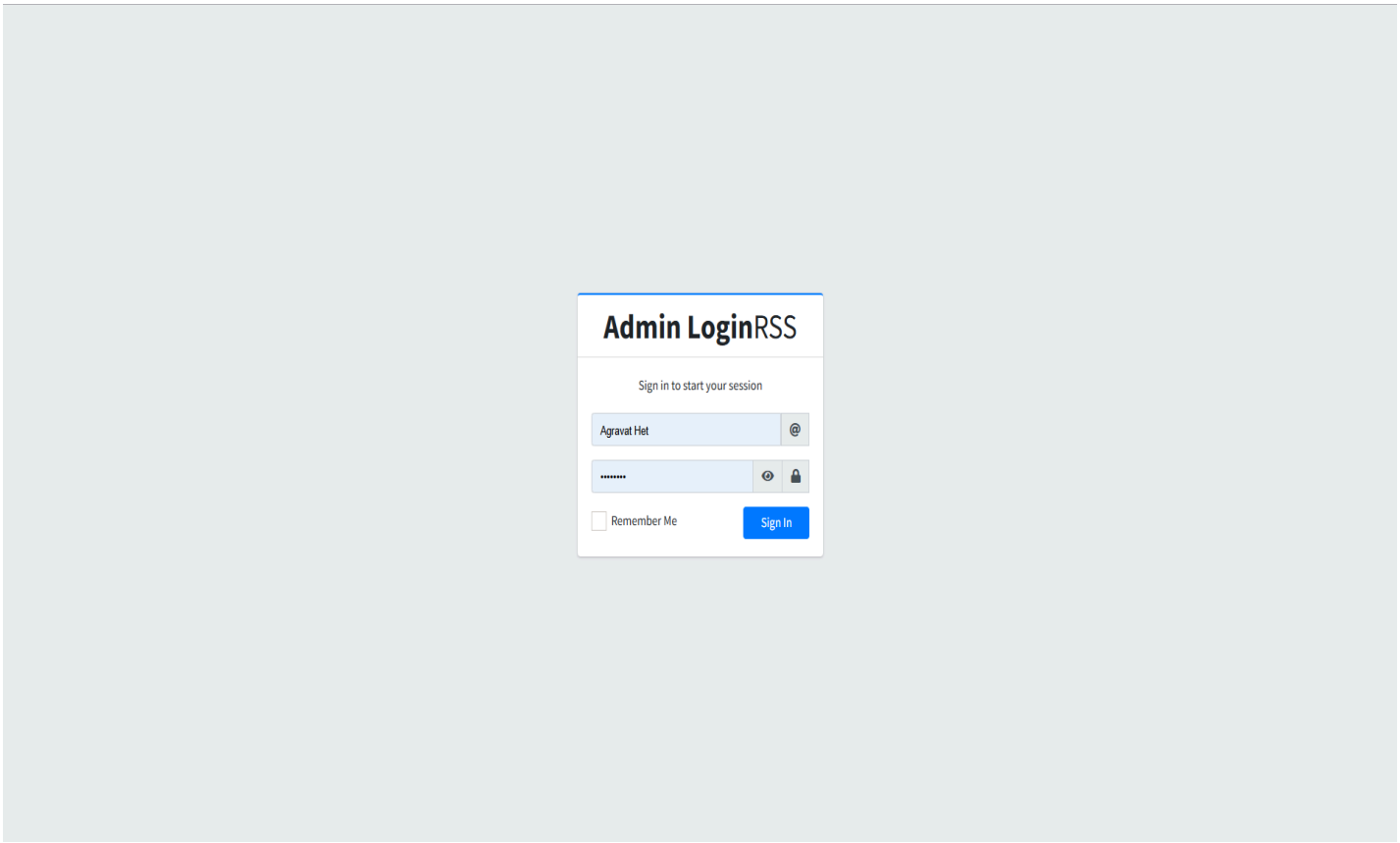
## Frontend Implementation:

- **HTML, CSS**: Used for structuring and styling the pages. HTML defined the structure, while CSS provided the design and aesthetics.
- **JavaScript & AJAX**: JavaScript was used to handle client-side interactions, including form validation and making asynchronous requests to the server using AJAX.
- **Bootstrap**: Bootstrap was used to ensure that the system was fully responsive, enhancing its usability on various devices, including desktops and mobile phones.

# Admin Side Screenshots:

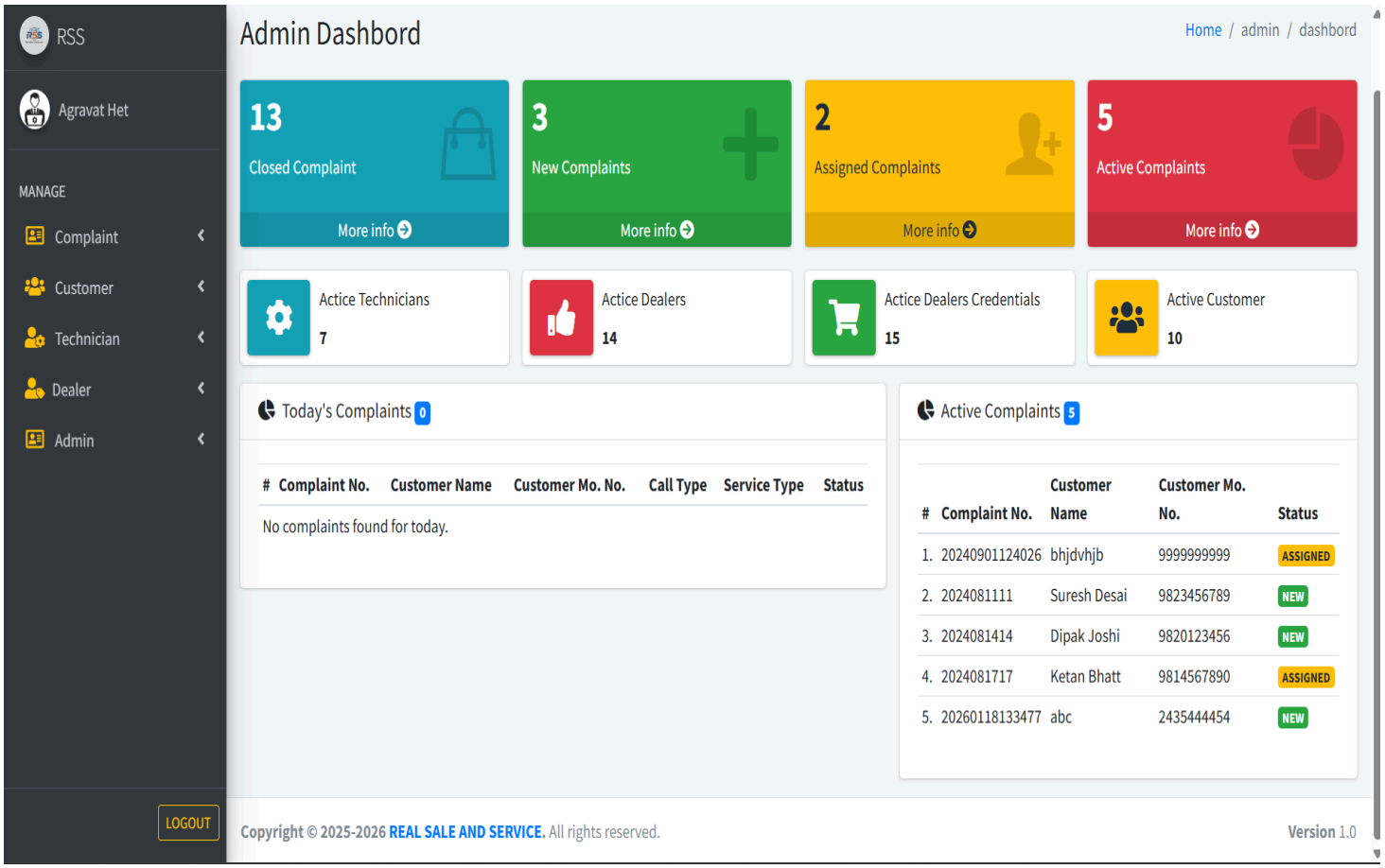
## 1. Login Page

- **Description**: The login page allows both **Admins** to access their respective dashboards. The form includes client-side validation and AJAX-based submission.
- **Screenshot**:



## 2. Admin Dashboard

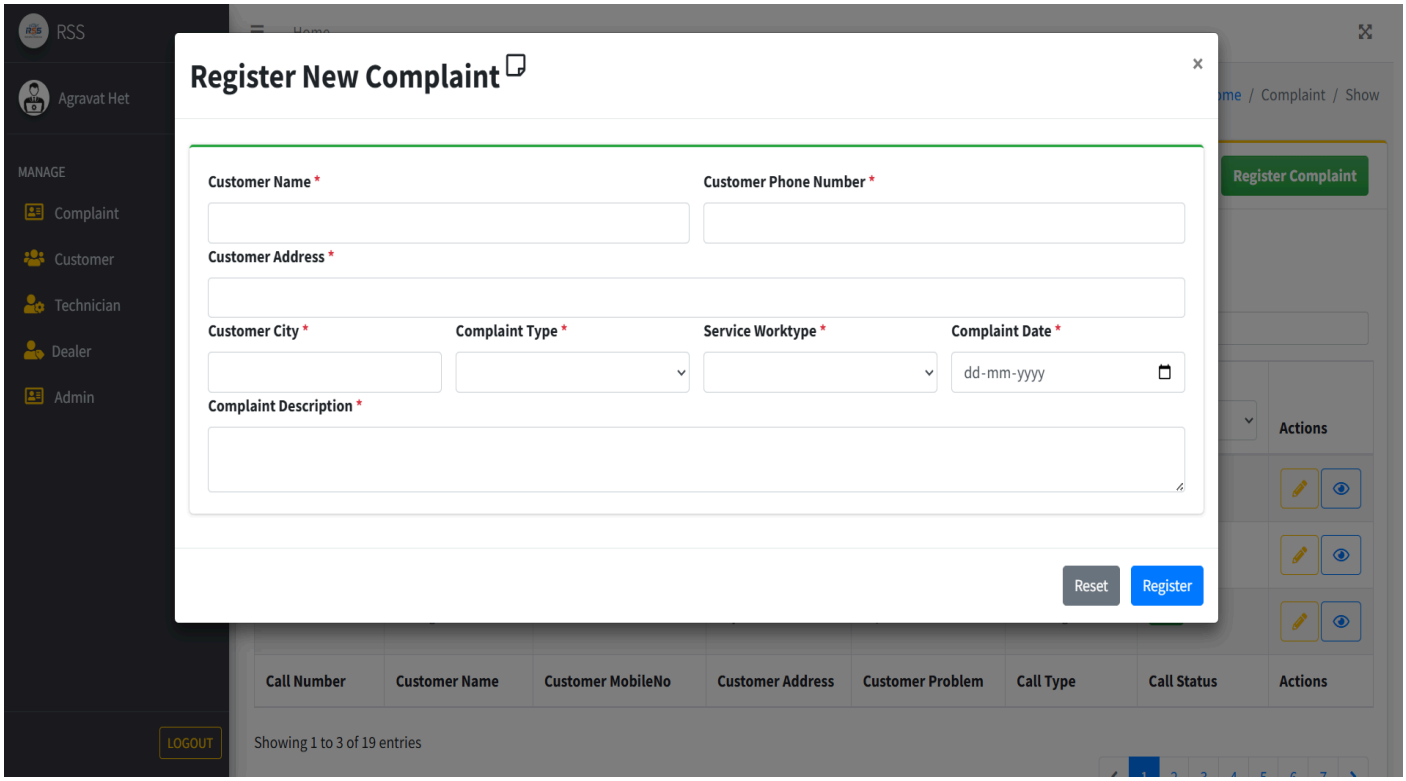
- **Description**: The Admin dashboard provides an overview of the system's current state, including:
    - Complaint statistics (New, Assigned, Closed).
    - The number of active technicians and dealers.
    - A list of recent complaints.
    - The total number of complaints, active customers, technicians, and dealers.
- **Screenshot**:

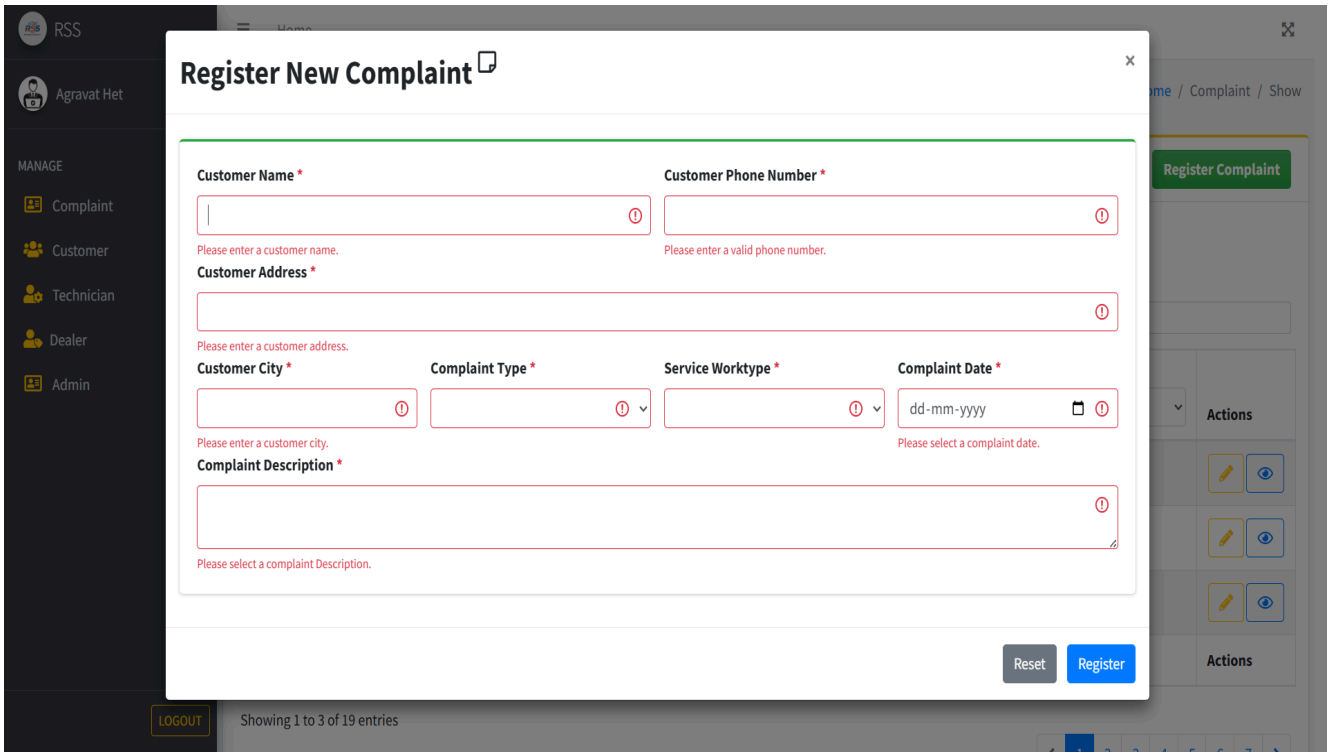


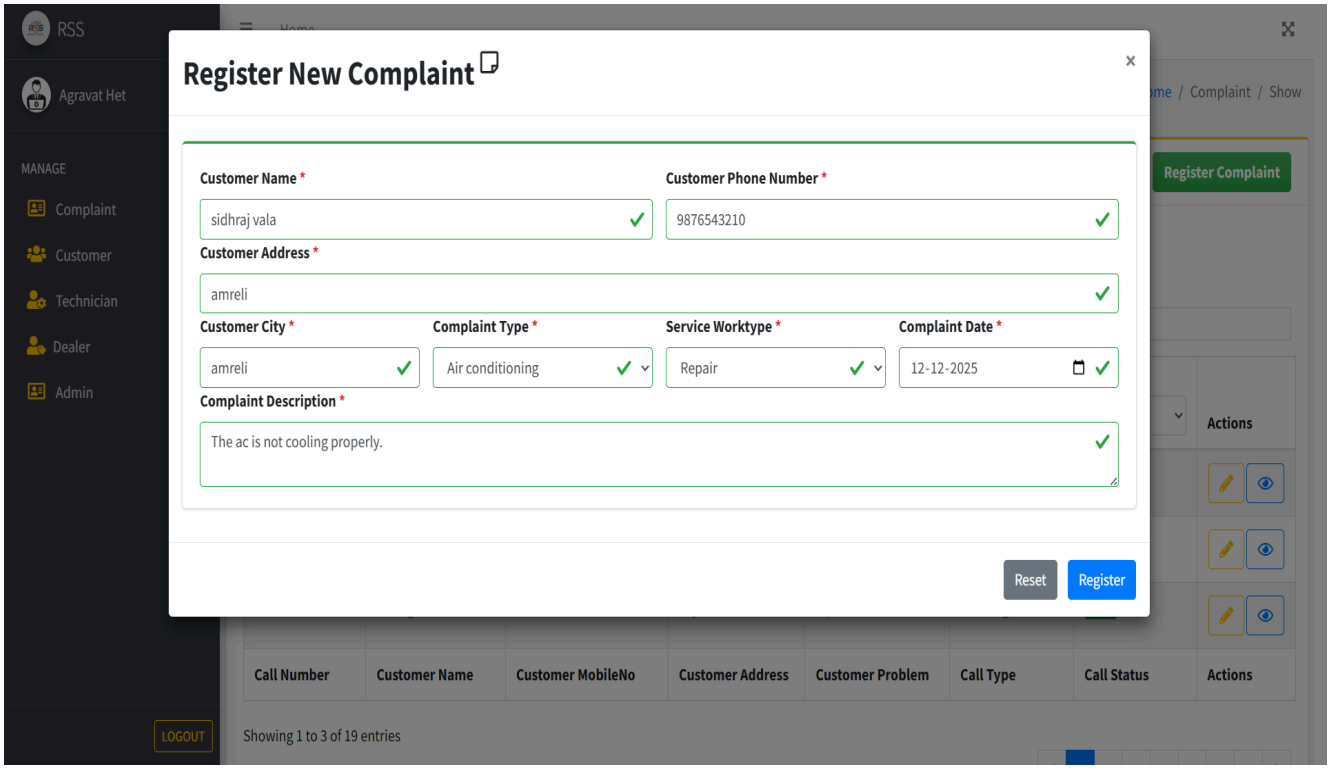
## 3. Complaint Management

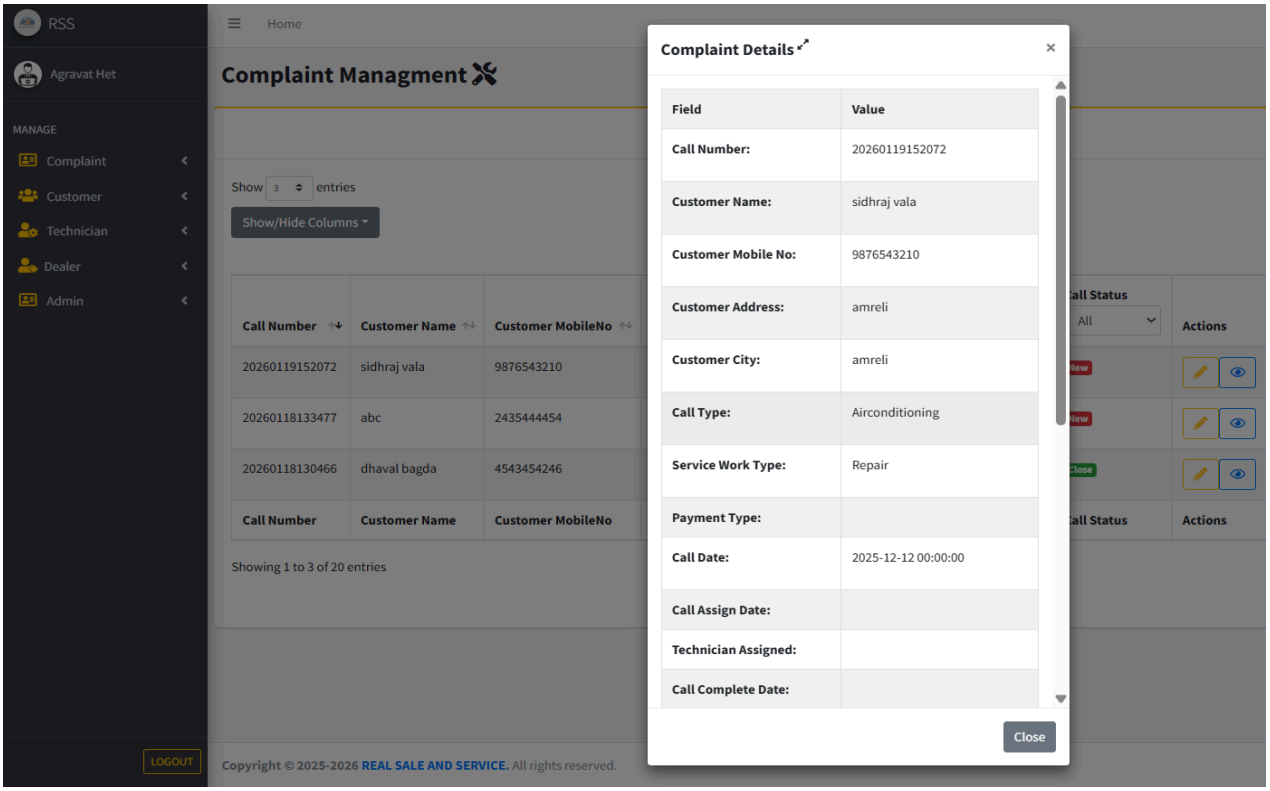
- **Description**: In this section, Admin can manage complaints registered by Dealers. Complaints can be inserted, viewed, updated, and assigned to technicians. This system streamlines the complaint resolution process, ensuring each complaint is handled efficiently and tracked throughout its lifecycle.
- **Operations**:
    - **Insert**: Admin or Dealers can register new complaints, providing customer and appliance details.
    - **Show**: A list of all complaints is displayed, including complaint ID, status, and assigned technician.
    - **Update**: Admin can update the complaint's status (assigned, closed), assign technicians, or cancel complaints.

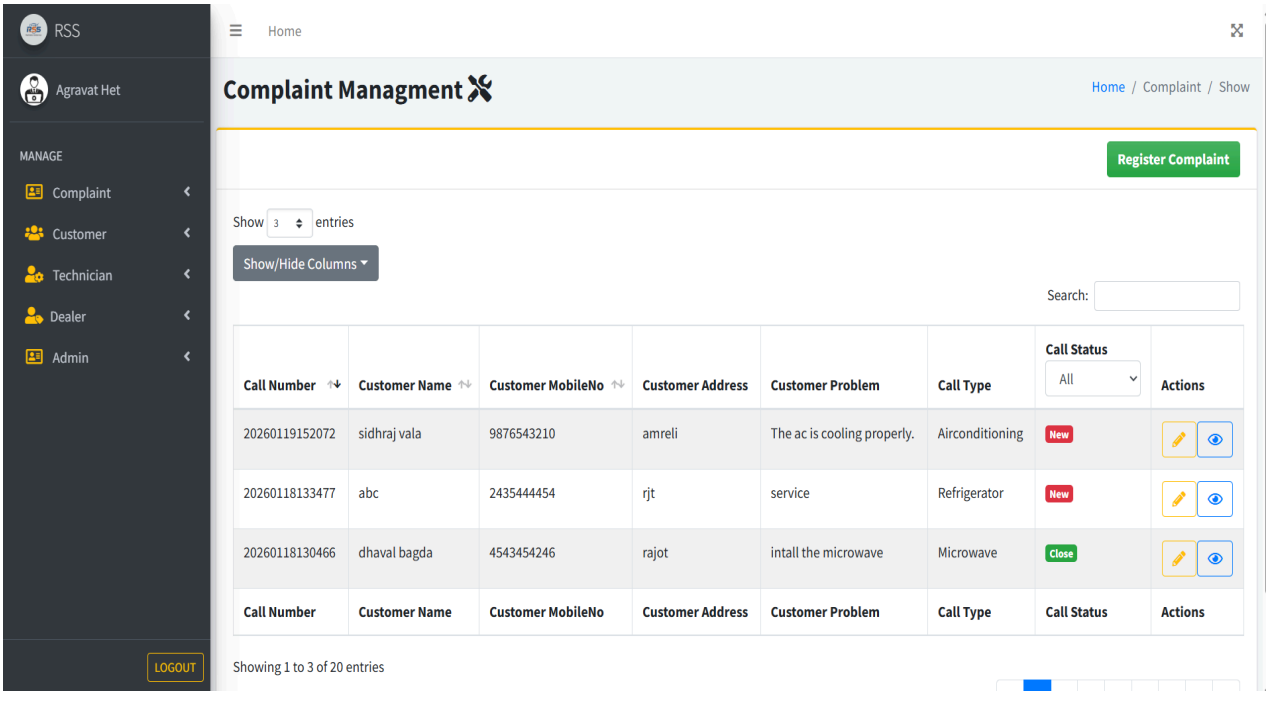
- **Screenshot**:

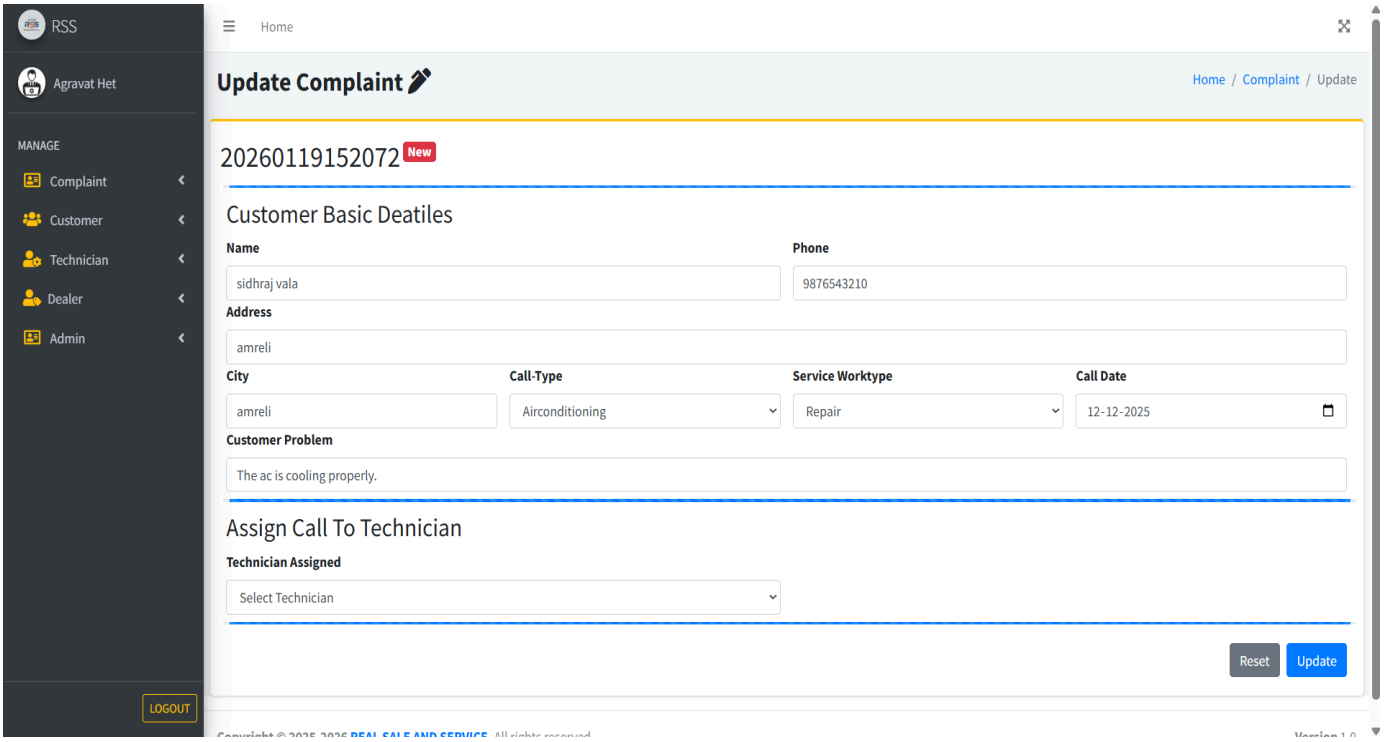



RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Register New Complaint
Customer Name *
Please enter a customer name.
Customer Phone Number *
Please enter a valid phone number.
Customer Address *
Please enter a customer address.
Customer City *
Please enter a customer city.
Complaint Type *
Service Worktype *
Complaint Date *
dd-mm-yyyy
Please select a complaint date.
Complaint Description *
Please select a complaint Description.
Reset
Register
Register Complaint
Actions
Showing 1 to 3 of 19 entries


RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Register New Complaint
Customer Name *
sidhraj vala
Customer Phone Number *
9876543210
Customer Address *
amreli
Customer City *
amreli
Complaint Type *
Air conditioning
Service Worktype *
Repair
Complaint Date *
12-12-2025
Complaint Description *
The ac is not cooling properly.
Reset
Register
Register Complaint
Actions
Call Number
Customer Name
Customer MobileNo
Customer Address
Customer Problem
Call Type
Call Status
Actions
Showing 1 to 3 of 19 entries

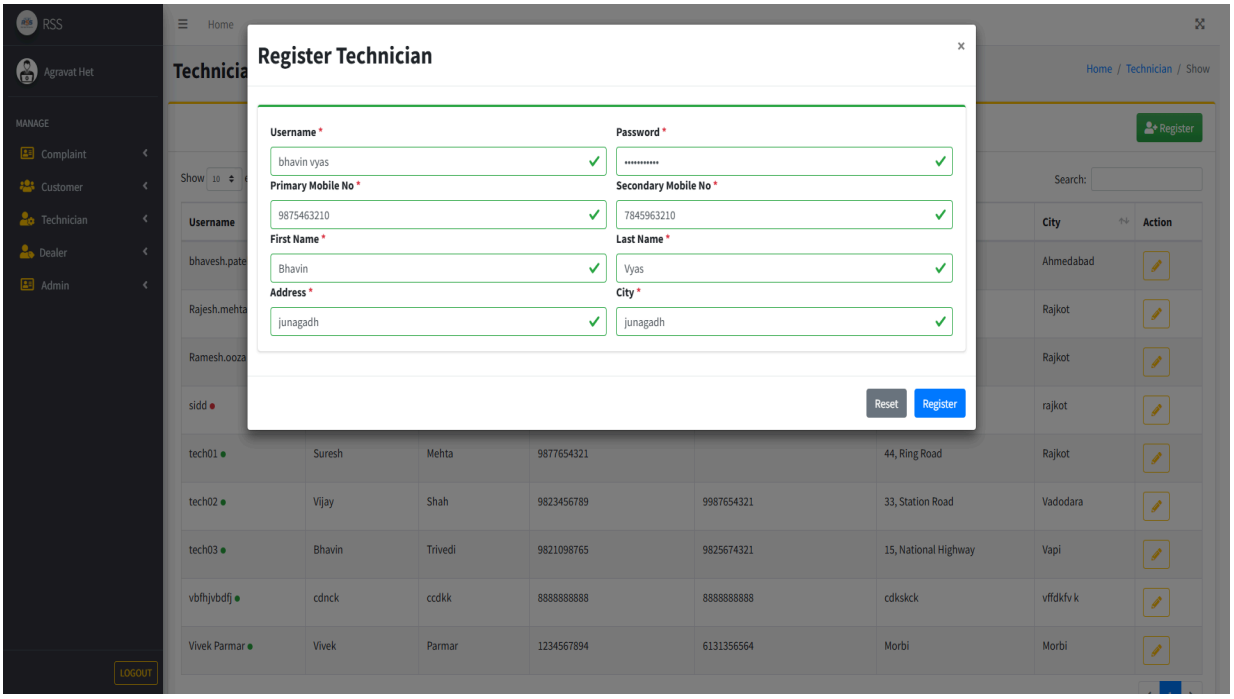

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Complaint Managment
Show 3 entries
Show/Hide Columns
Call Number
Customer Name
Customer MobileNo
20260119152072
sidhraj vala
9876543210
20260118133477
abc
2435444454
20260118130466
dhaval bagda
4543454246
Showing 1 to 3 of 20 entries
Actions
Complaint Details
Field
Value
Call Number:
20260119152072
Customer Name:
sidhraj vala
Customer Mobile No:
9876543210
Customer Address:
amreli
Customer City:
amreli
Call Type:
Airconditioning
Service Work Type:
Repair
Payment Type:
Call Date:
2025-12-12 00:00:00
Call Assign Date:
Technician Assigned:
Call Complete Date:
Close
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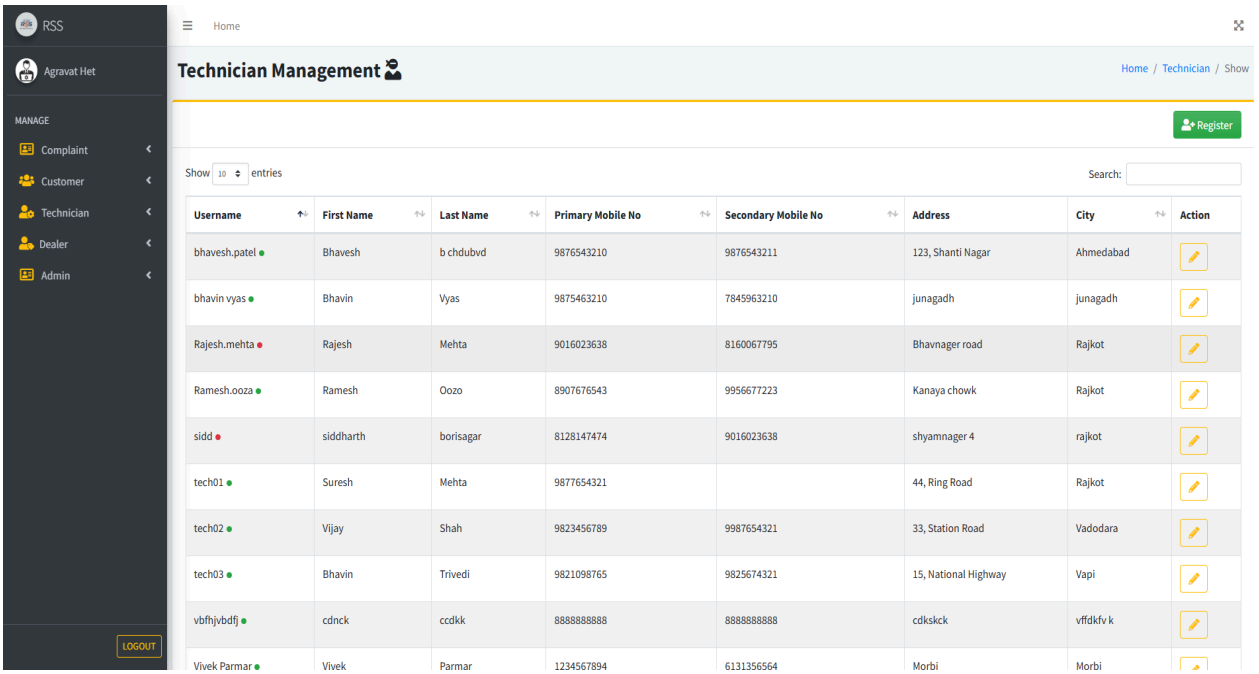

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Complaint Managment
Home / Complaint / Show
Register Complaint
Show 3 entries
Show/Hide Columns
Search:
Call Number
Customer Name
Customer MobileNo
Customer Address
Customer Problem
Call Type
Call Status
All
Actions
20260119152072
sidhraj vala
9876543210
amreli
The ac is cooling properly.
Airconditioning
New
20260118133477
abc
2435444454
rjt
service
Refrigerator
New
20260118130466
dhaval bagda
4543454246
rajot
intall the microwave
Microwave
Close
Showing 1 to 3 of 20 entries

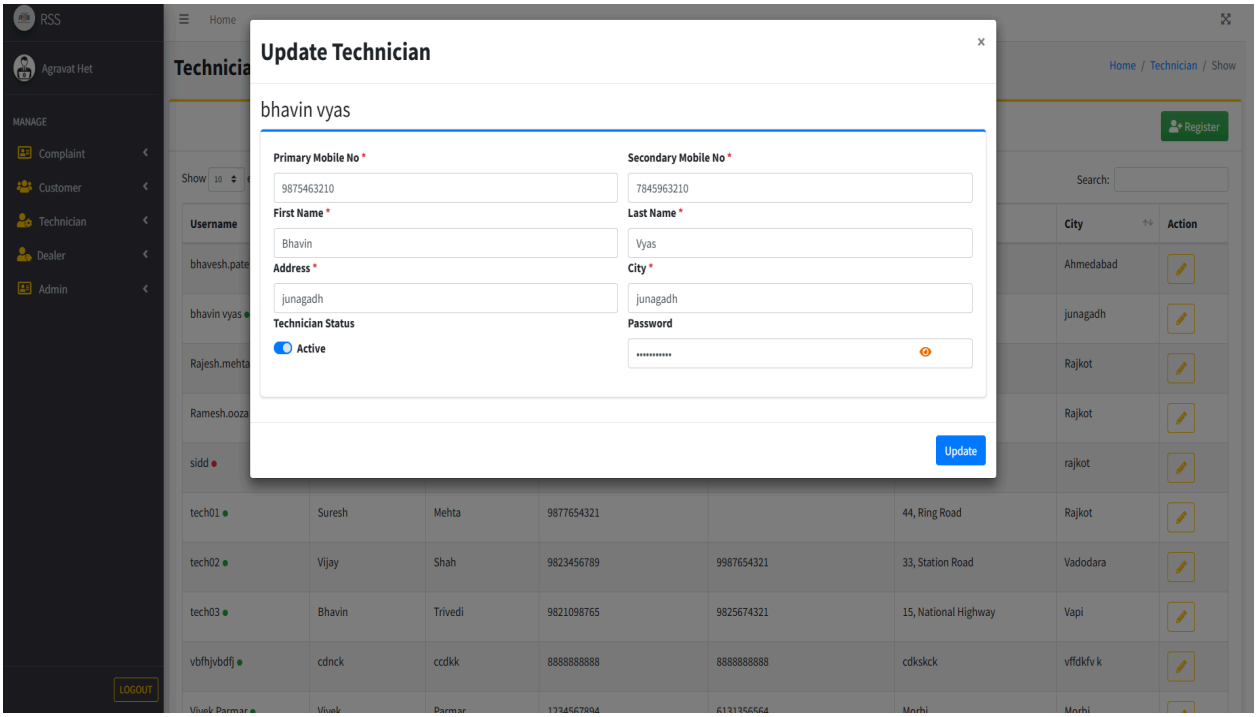

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Update Complaint
Home / Complaint / Update
20260119152072
New
Customer Basic Deatiles
Name
sidhraj vala
Phone
9876543210
Address
amreli
City
amreli
Call-Type
Airconditioning
Service Worktype
Repair
Call Date
12-12-2025
Customer Problem
The ac is cooling properly.
Assign Call To Technician
Technician Assigned
Select Technician
Reset
Update

## 4. Technician Management

- **Description**: This interface allows Admins to manage technicians, including adding new technicians, updating their status, and assigning them to complaints.
- **Operations**:
    - **Insert**: Add new technicians, including availability.
    - **Show**: View the list of technicians along with their current status (active/inactive).
    - **Update**: Modify technician details such as their availability, or contact information.
- **Screenshot**:

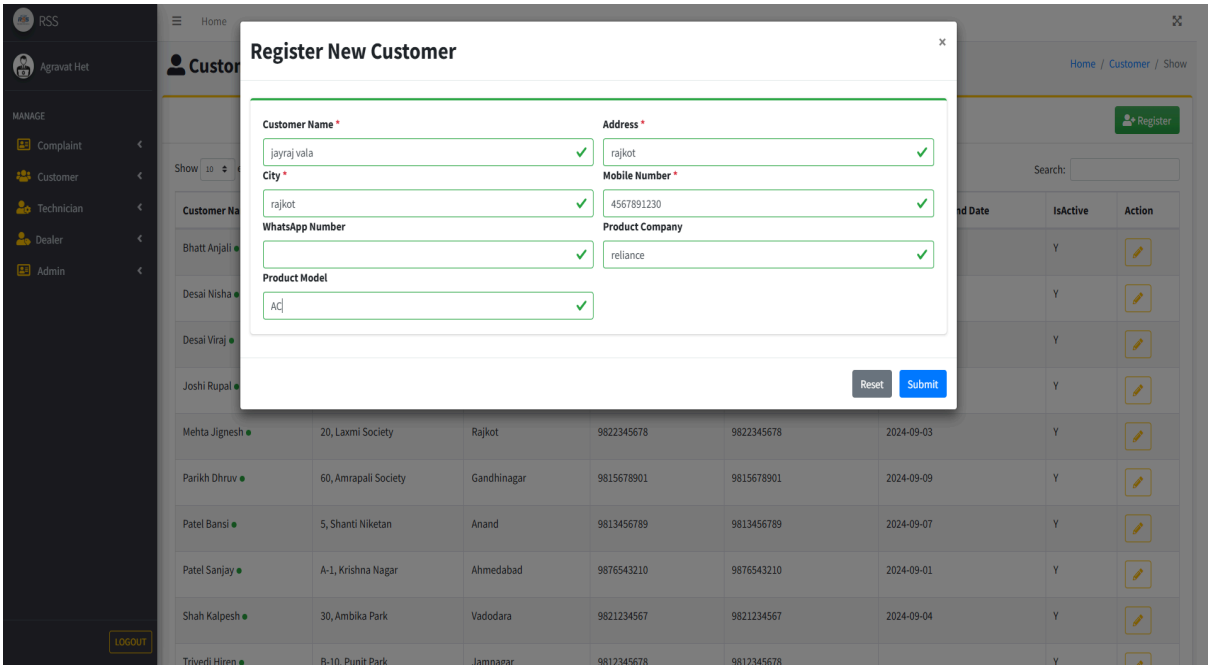


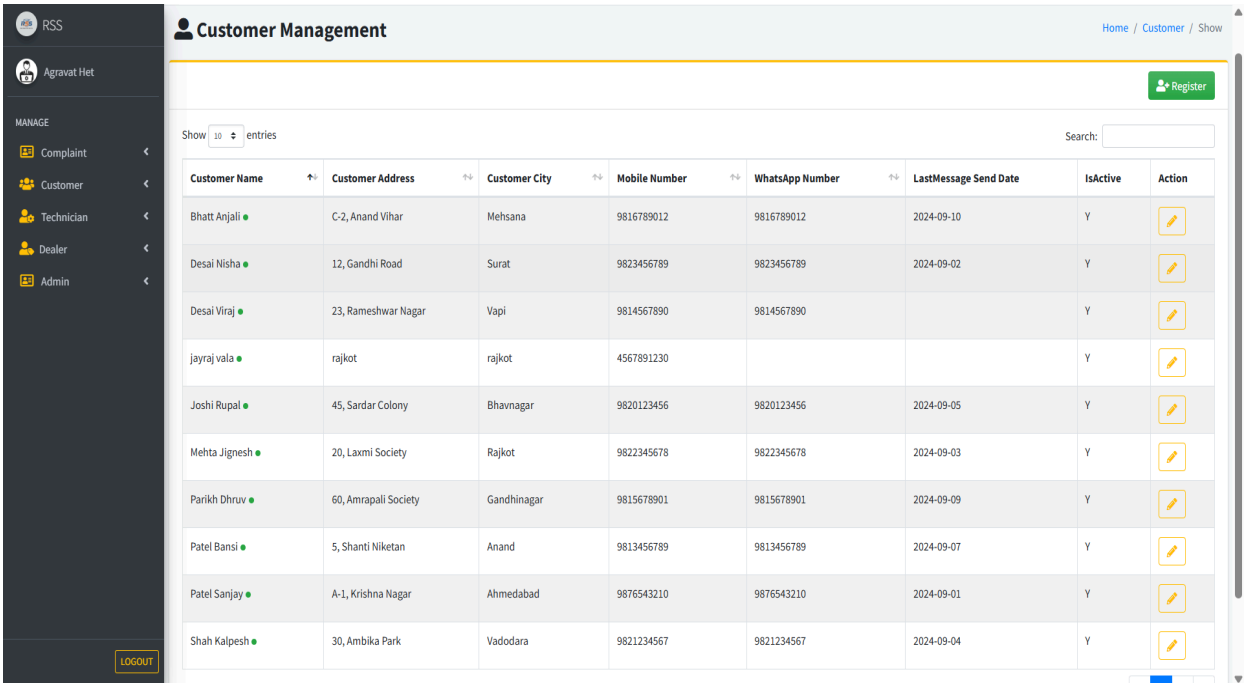

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Technician Management
Home / Technician / Show
Register
Show 10 entries
Search:
Username
First Name
Last Name
Primary Mobile No
Secondary Mobile No
Address
City
Action
bhavesh.patel
Bhavesh
b chdubvd
9876543210
9876543211
123, Shanti Nagar
Ahmedabad
bhavin vyas
Bhavin
Vyas
9875463210
7845963210
junagadh
junagadh
Rajesh.mehta
Rajesh
Mehta
9016023638
8160067795
Bhavnager road
Rajkot
Ramesh.ooza
Ramesh
Oozo
8907676543
9956677223
Kanaya chowk
Rajkot
sidd
siddharth
borisagar
8128147474
9016023638
shyamnager 4
rajkot
tech01
Suresh
Mehta
9877654321
44, Ring Road
Rajkot
tech02
Vijay
Shah
9823456789
9987654321
33, Station Road
Vadodara
tech03
Bhavin
Trivedi
9821098765
9825674321
15, National Highway
Vapi
vbfhjvbdfj
cdnck
ccdkk
8888888888
8888888888
cdkskck
vffdkfv k
Vivek Parmar
Vivek
Parmar
1234567894
6131356564
Morbi
Morbi

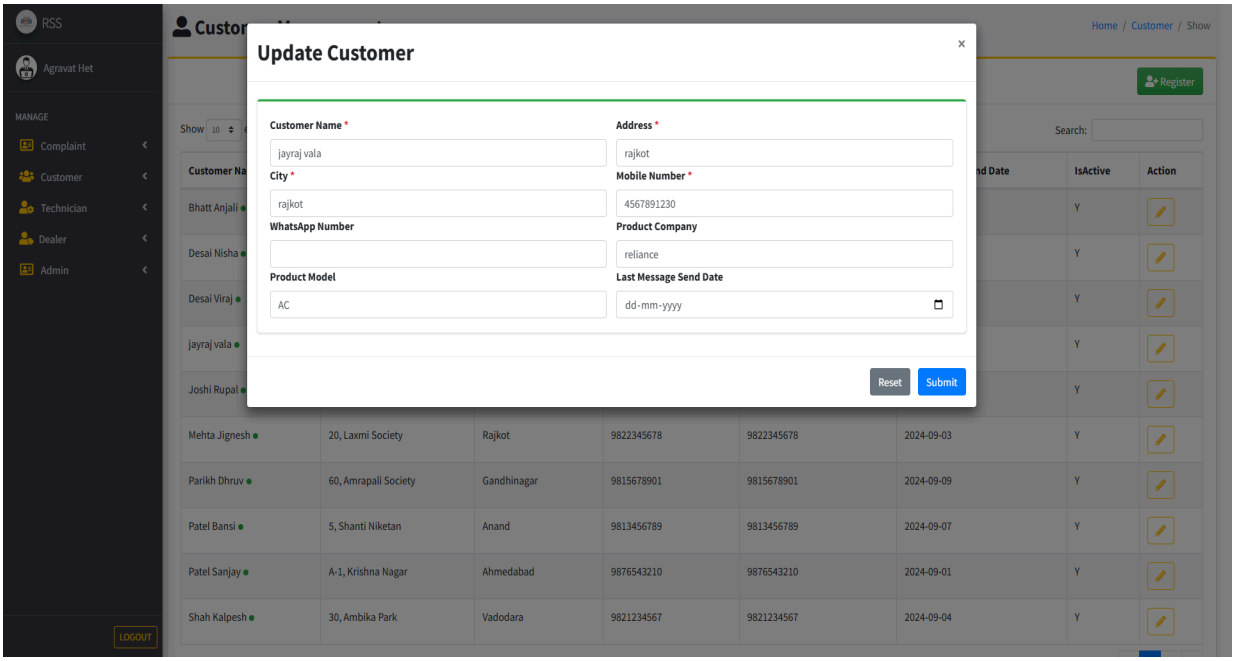

Update Technician
bhavin vyas
Primary Mobile No *
9875463210
Secondary Mobile No *
7845963210
First Name *
Bhavin
Last Name *
Vyas
Address *
junagadh
City *
junagadh
Technician Status
Active
Password
Update

## 5. CustomerManagement

- **Description**: The Customer Management module allows the Admin to manage customer information. This includes adding new customers, viewing existing customer details, and updating their information. This ensures the database remains accurate and up-to-date for better service tracking.
- **Operations**:
    - **Insert**: Add new customers with personal and contact details.
    - **Show**: Display a table of all registered customers with options to filter by name.
    - **Update**: Modify customer information such as name, address, or contact number when needed.
- **Screenshot**:

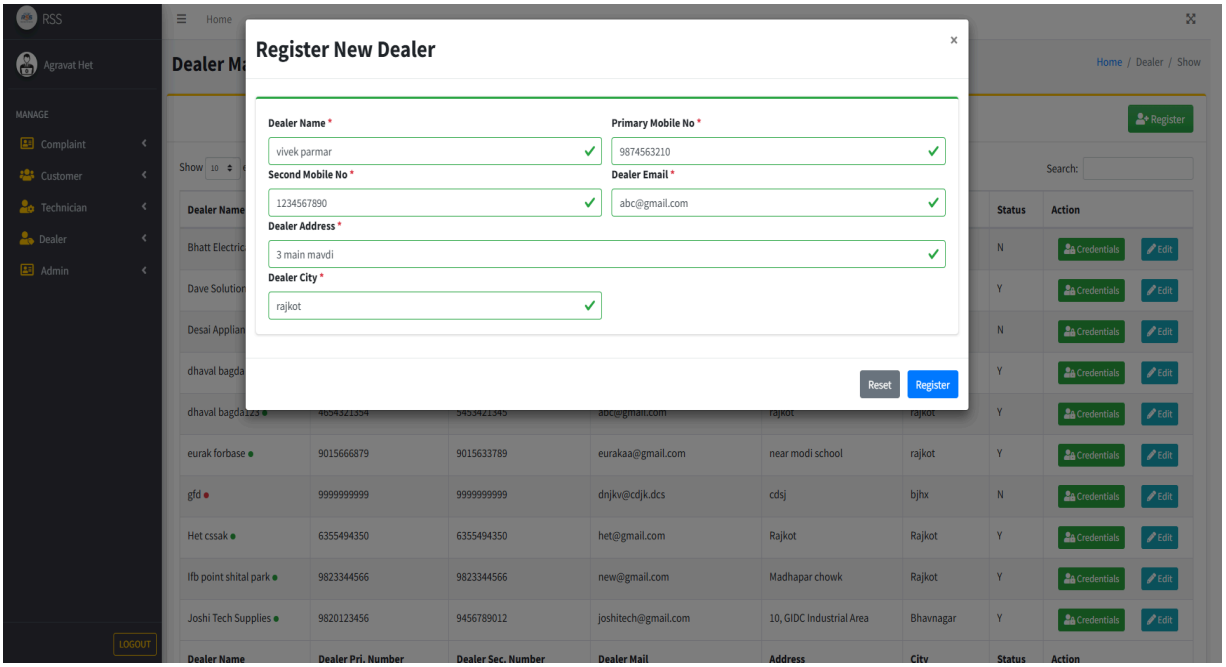


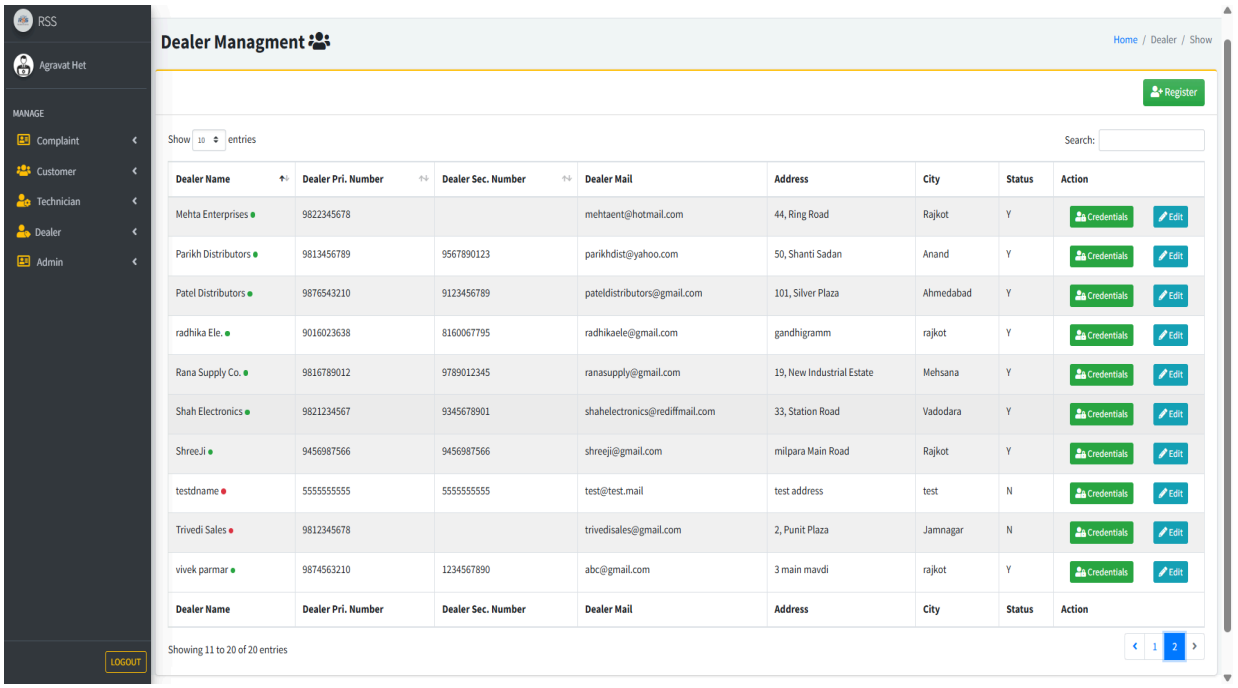

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Customer Management
Home / Customer / Show
Register
Show 10 entries
Search:
Customer Name
Customer Address
Customer City
Mobile Number
WhatsApp Number
LastMessage Send Date
IsActive
Action
Bhatt Anjali
C-2, Anand Vihar
Mehsana
9816789012
9816789012
2024-09-10
Y
Desai Nisha
12, Gandhi Road
Surat
9823456789
9823456789
2024-09-02
Y
Desai Viraj
23, Rameshwar Nagar
Vapi
9814567890
9814567890
Y
jayraj vala
rajkot
rajkot
4567891230
Y
Joshi Rupal
45, Sardar Colony
Bhavnagar
9820123456
9820123456
2024-09-05
Y
Mehta Jignesh
20, Laxmi Society
Rajkot
9822345678
9822345678
2024-09-03
Y
Parikh Dhruv
60, Amrapali Society
Gandhinagar
9815678901
9815678901
2024-09-09
Y
Patel Bansi
5, Shanti Niketan
Anand
9813456789
9813456789
2024-09-07
Y
Patel Sanjay
A-1, Krishna Nagar
Ahmedabad
9876543210
9876543210
2024-09-01
Y
Shah Kalpesh
30, Ambika Park
Vadodara
9821234567
9821234567
2024-09-04
Y

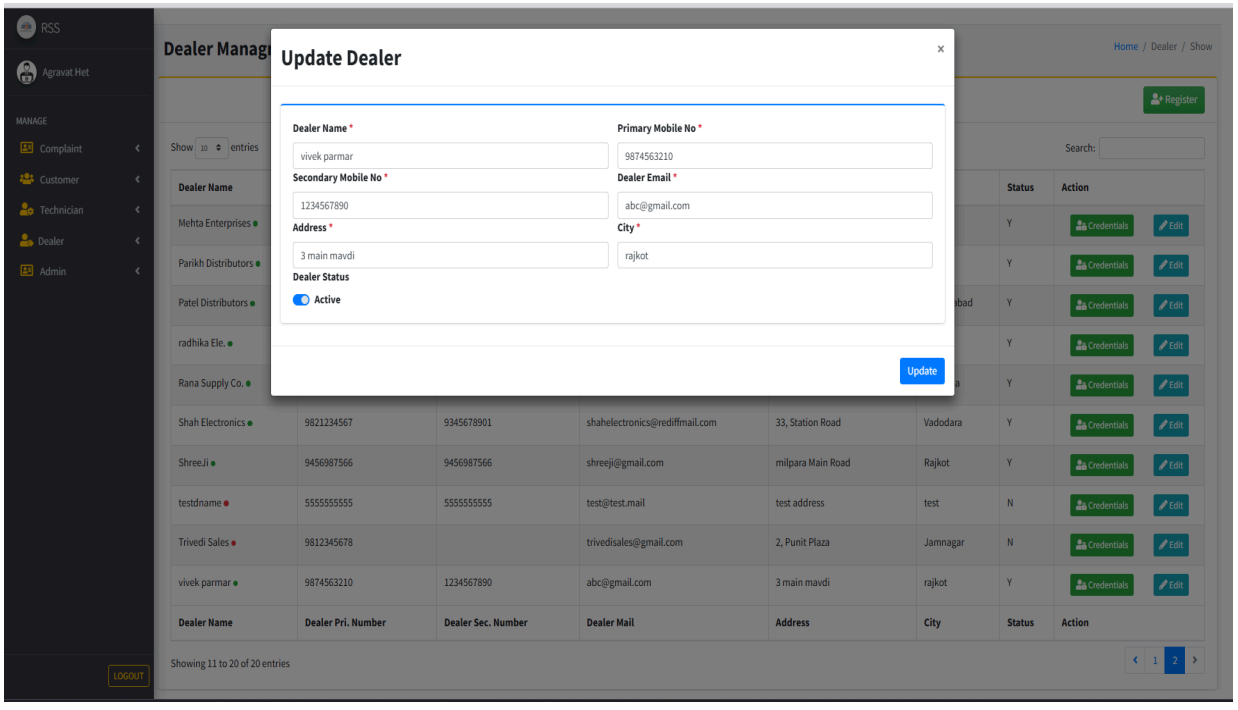

Update Customer
Customer Name *
jayraj vala
Address *
rajkot
City *
rajkot
Mobile Number *
4567891230
WhatsApp Number
Product Company
reliance
Product Model
AC
Last Message Send Date
dd-mm-yyyy
Reset
Submit

## 6. Dealer Management

- **Description**: Dealer Management allows Admin to maintain records of all registered dealers. Dealers are responsible for registering complaints on behalf of customers. This section helps Admin ensure that dealer details are accurate and updated.
- **Operations**:
    - **Insert**: Register new dealers, providing company and contact details.
    - **Show**: Display the list of dealers, along with their active status.
    - **Update**: Update dealer information and control their system access (activate/deactivate).
- **Screenshot**:

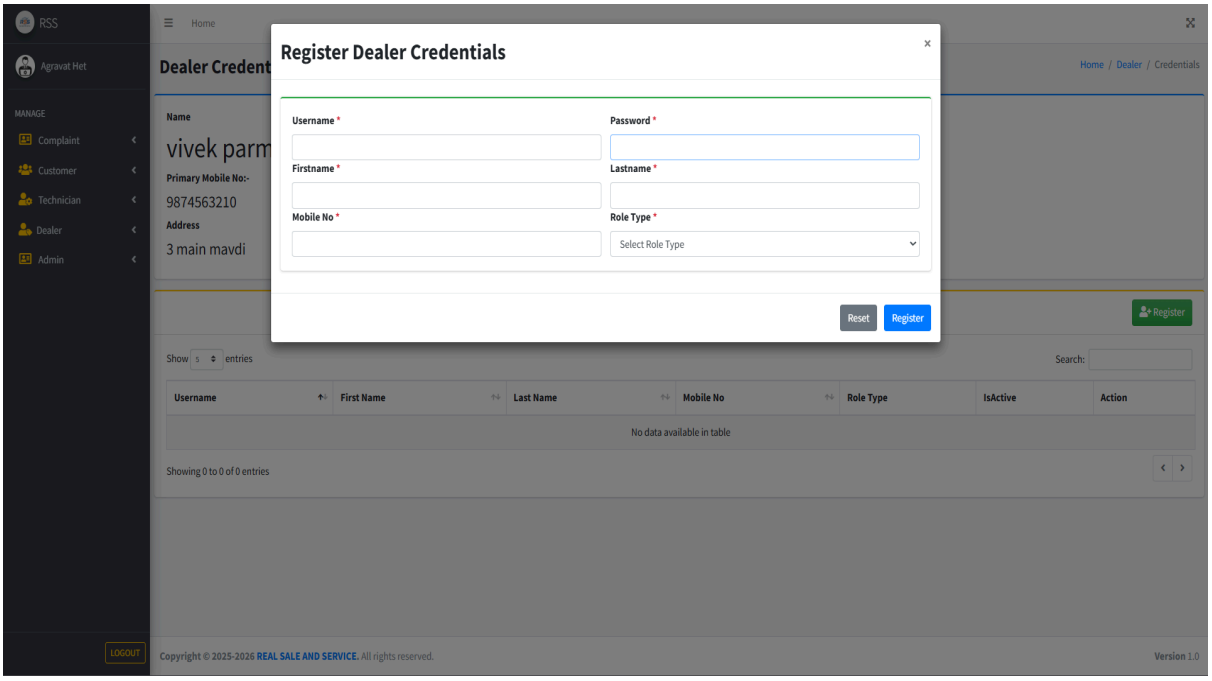


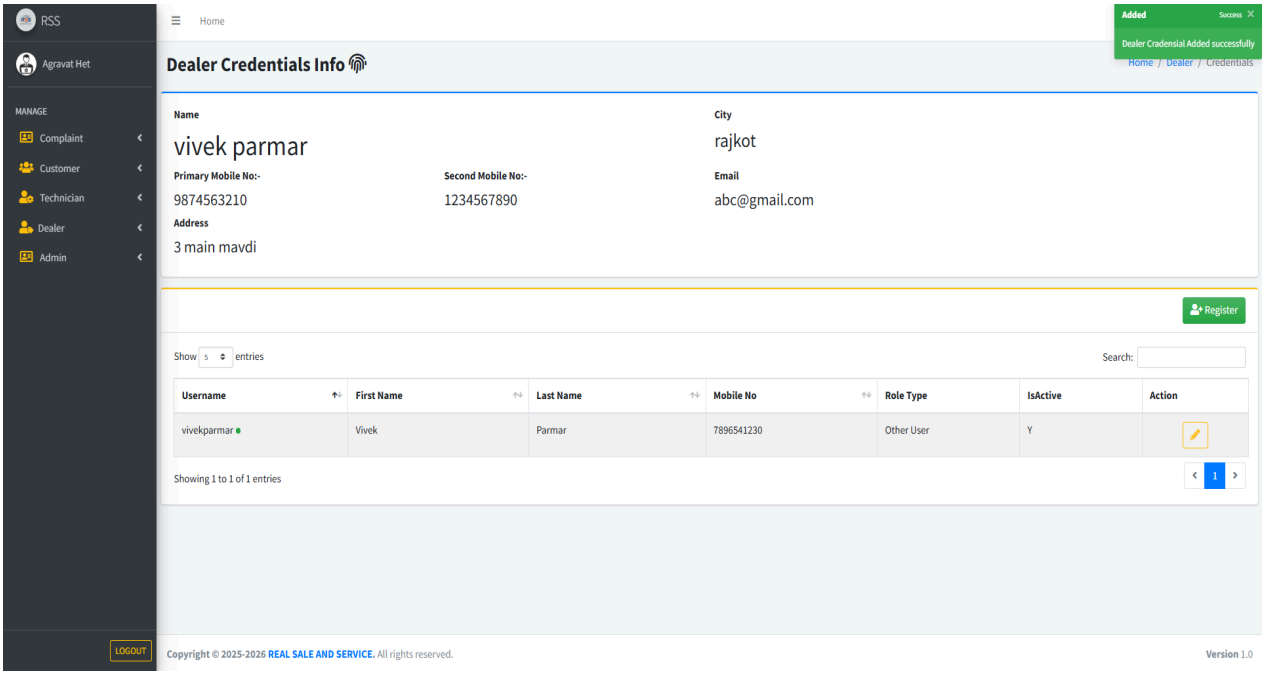

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Dealer Managment
Home / Dealer / Show
Register
Show 10 entries
Search:
Dealer Name
Dealer Pri. Number
Dealer Sec. Number
Dealer Mail
Address
City
Status
Action
Mehta Enterprises 9822345678 mehtaent@hotmail.com 44, Ring Road Rajkot Y
Parikh Distributors 9813456789 9567890123 parikhdist@yahoo.com 50, Shanti Sadan Anand Y
Patel Distributors 9876543210 9123456789 pateldistributors@gmail.com 101, Silver Plaza Ahmedabad Y
radhika Ele. 9016023638 8160067795 radhikaele@gmail.com gandhigramm rajkot Y
Rana Supply Co. 9816789012 9789012345 ranasupply@gmail.com 19, New Industrial Estate Mehsana Y
Shah Electronics 9821234567 9345678901 shahelectronics@rediffmail.com 33, Station Road Vadodara Y
ShreeJi 9456987566 9456987566 shreeji@gmail.com milpara Main Road Rajkot Y
testdname 5555555555 5555555555 test@test.mail test address test N
Trivedi Sales 9812345678 trivedisales@gmail.com 2, Punit Plaza Jamnagar N
vivek parmar 9874563210 1234567890 abc@gmail.com 3 main mavdi rajkot Y
Credentials
Edit
Showing 11 to 20 of 20 entries
1
2

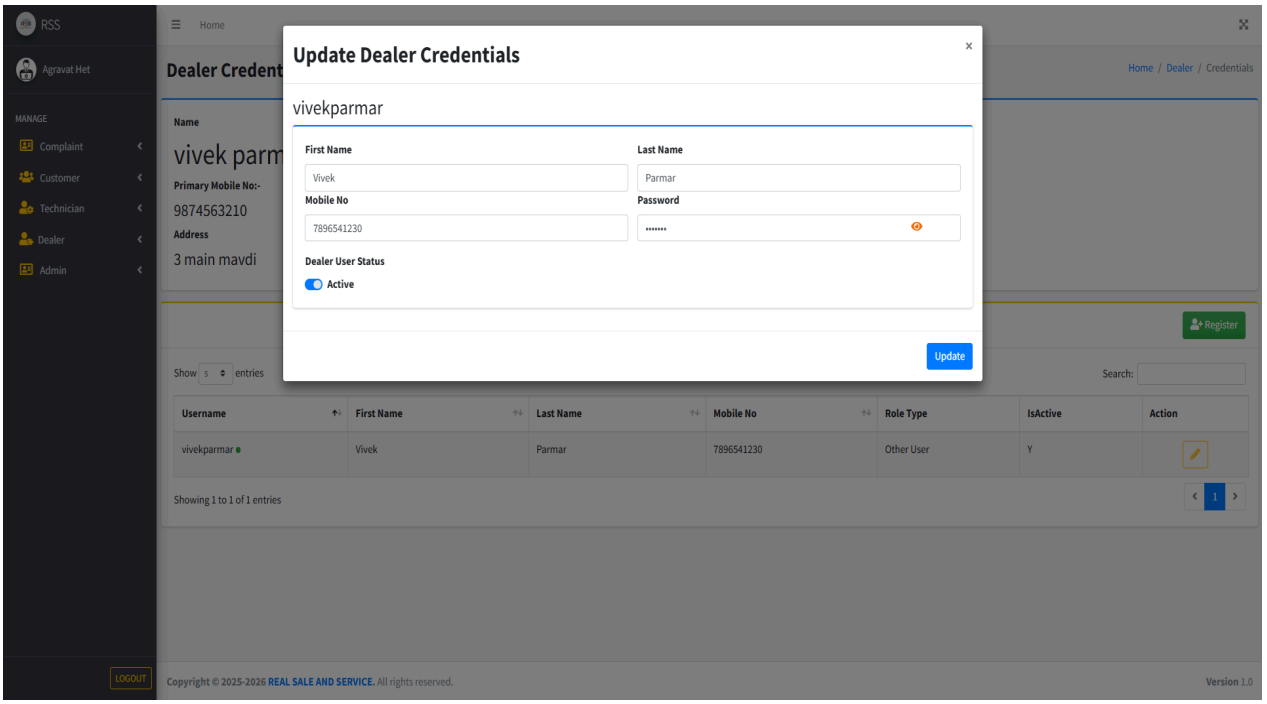

Update Dealer
Dealer Name *
vivek parmar
Primary Mobile No *
9874563210
Secondary Mobile No *
1234567890
Dealer Email *
abc@gmail.com
Address *
3 main mavdi
City *
rajkot
Dealer Status
Active
Update

## 7. Dealer Credentials Management

- **Description**: This section focuses on managing login credentials for dealers. Admin can create and update login details, ensuring that dealers have secure access to the system. The credentials are necessary for dealers to submit complaints and track service requests.
- **Operations**:
    - **Insert**: Create login credentials for new dealers upon registration.
    - **Show**: Display dealer login information (excluding sensitive password data).
    - **Update**: Admin can reset dealer passwords or deactivate login access if needed.
- **Screenshot**:



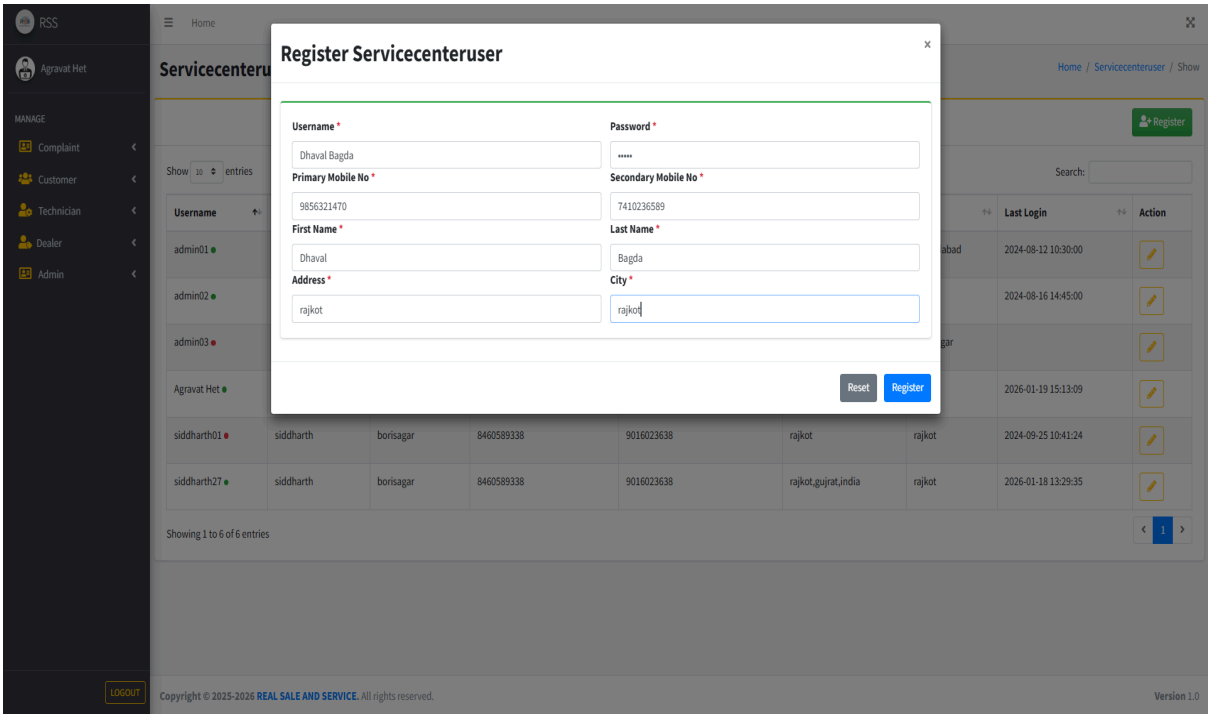

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Dealer Credentials Info
Added
Success
Dealer Cradensial Added successfully
Home / Dealer / Credentials
Name
vivek parmar
City
rajkot
Primary Mobile No:-
9874563210
Second Mobile No:-
1234567890
Email
abc@gmail.com
Address
3 main mavdi
Register
Show 5 entries
Search:
Username
First Name
Last Name
Mobile No
Role Type
IsActive
Action
vivekparmar
Vivek
Parmar
7896541230
Other User
Y
Showing 1 to 1 of 1 entries
1
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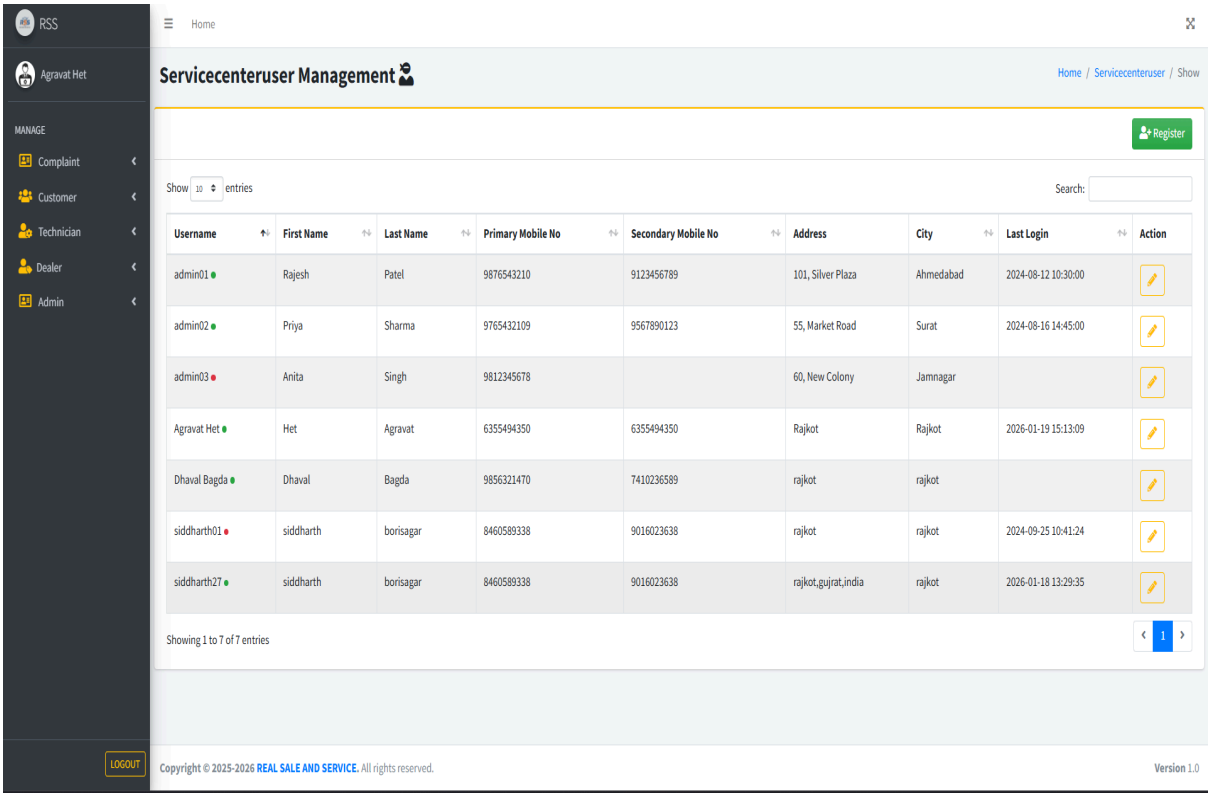

Update Dealer Credentials
vivekparmar
First Name
Vivek
Last Name
Parmar
Mobile No
7896541230
Password
Dealer User Status
Active
Update

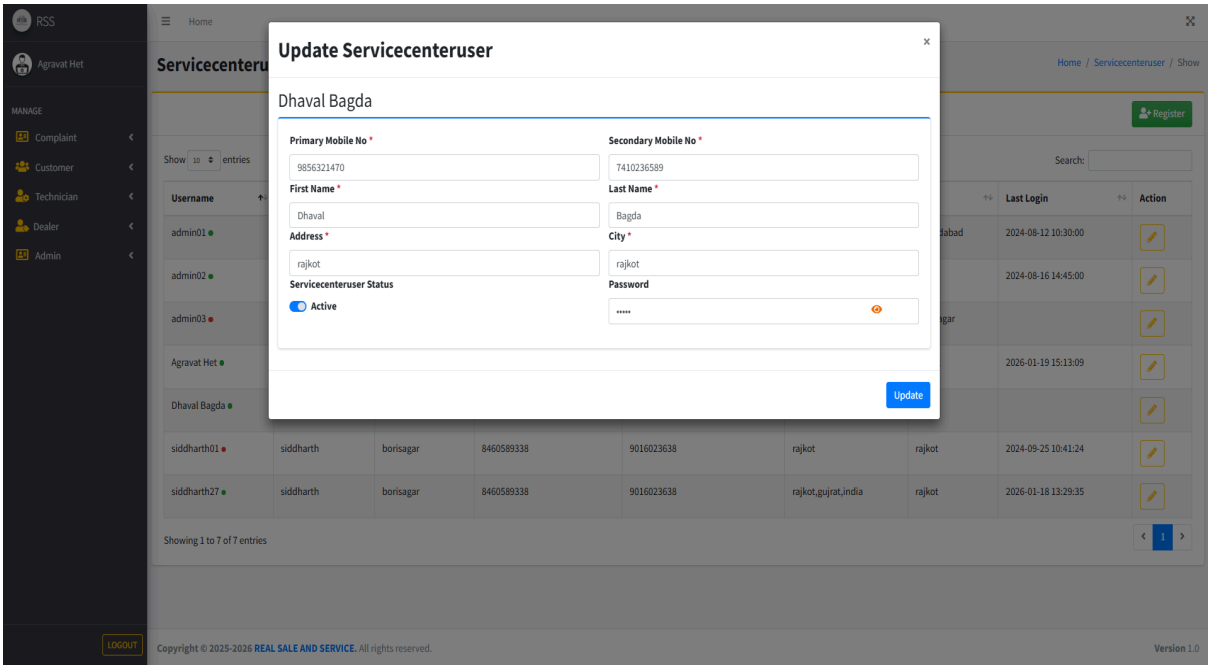
## 8. Admin Credentials Management (Service Centre User)

- **Description**: Admin Credentials Management ensures that all Admin users of the SCMS system have secure access. The system allows admins to manage the credentials of other admin users, ensuring only authorized personnel can access sensitive service center data.
- **Operations**:
    - **Insert**: Create login credentials for new Admin users.
    - **Show**: Display a list of all Admin users and their last login activity.
    - **Update**: Reset passwords of Admin users.

- **Screenshot**:



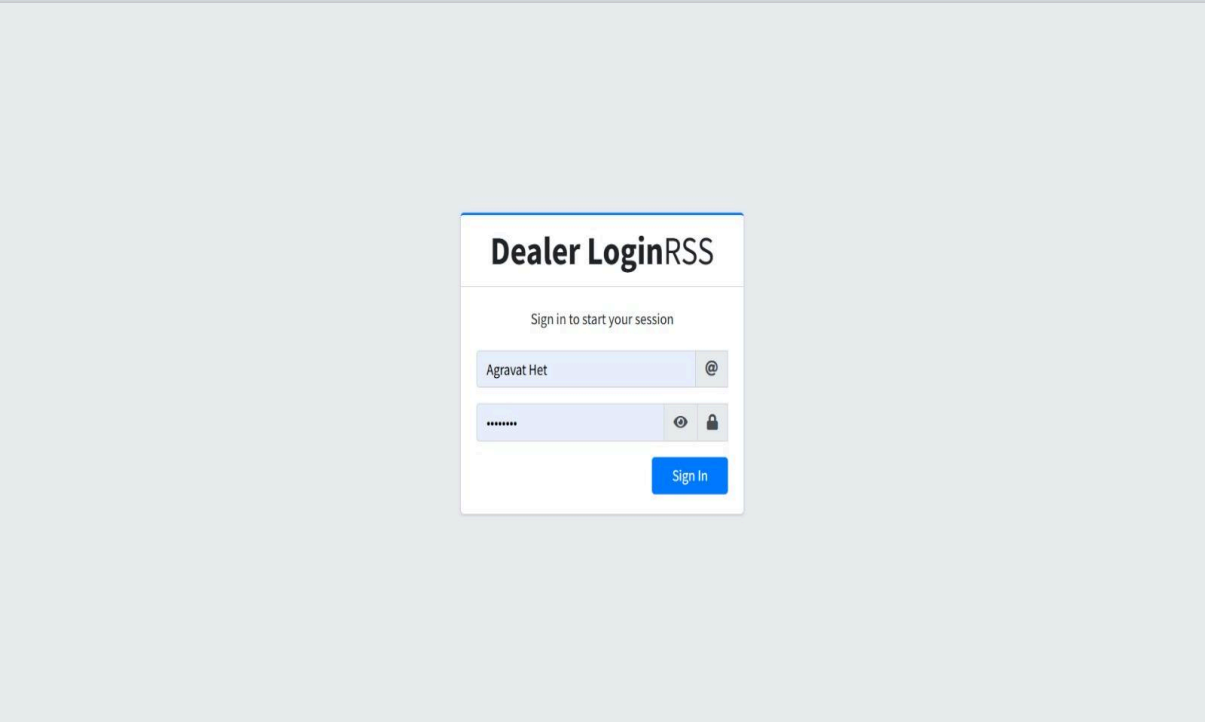

RSS
Agravat Het
MANAGE
Complaint
Customer
Technician
Dealer
Admin
LOGOUT
Home
Servicecenteruser Management
Home / Servicecenteruser / Show
Register
Show 10 entries
Search:
Username
First Name
Last Name
Primary Mobile No
Secondary Mobile No
Address
City
Last Login
Action
admin01 Rajesh Patel 9876543210 9123456789 101, Silver Plaza Ahmedabad 2024-08-12 10:30:00
admin02 Priya Sharma 9765432109 9567890123 55, Market Road Surat 2024-08-16 14:45:00
admin03 Anita Singh 9812345678 60, New Colony Jamnagar
Agravat Het Het Agravat 6355494350 6355494350 Rajkot Rajkot 2026-01-19 15:13:09
Dhaval Bagda Dhaval Bagda 9856321470 7410236589 rajkot rajkot
siddharth01 siddharth borisagar 8460589338 9016023638 rajkot rajkot 2024-09-25 10:41:24
siddharth27 siddharth borisagar 8460589338 9016023638 rajkot,gujrat,india rajkot 2026-01-18 13:29:35
Showing 1 to 7 of 7 entries
1
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Version 1.0


Update Servicecenteruser
Dhaval Bagda
Primary Mobile No *
9856321470
Secondary Mobile No *
7410236589
First Name *
Dhaval
Last Name *
Bagda
Address *
rajkot
City *
rajkot
Servicecenteruser Status
Active
Password
Update

# DealerSide Screenshots:

## 1. Login Page

- **Description**: The **Dealer Login** page provides a secure entry point for dealers to access the Service Center Management System (SCMS). Only authorized dealers with valid credentials can log in to manage complaints and track service requests. The login system ensures that each dealer’s data is protected and only accessible by authenticated users.
- **Screenshot**:

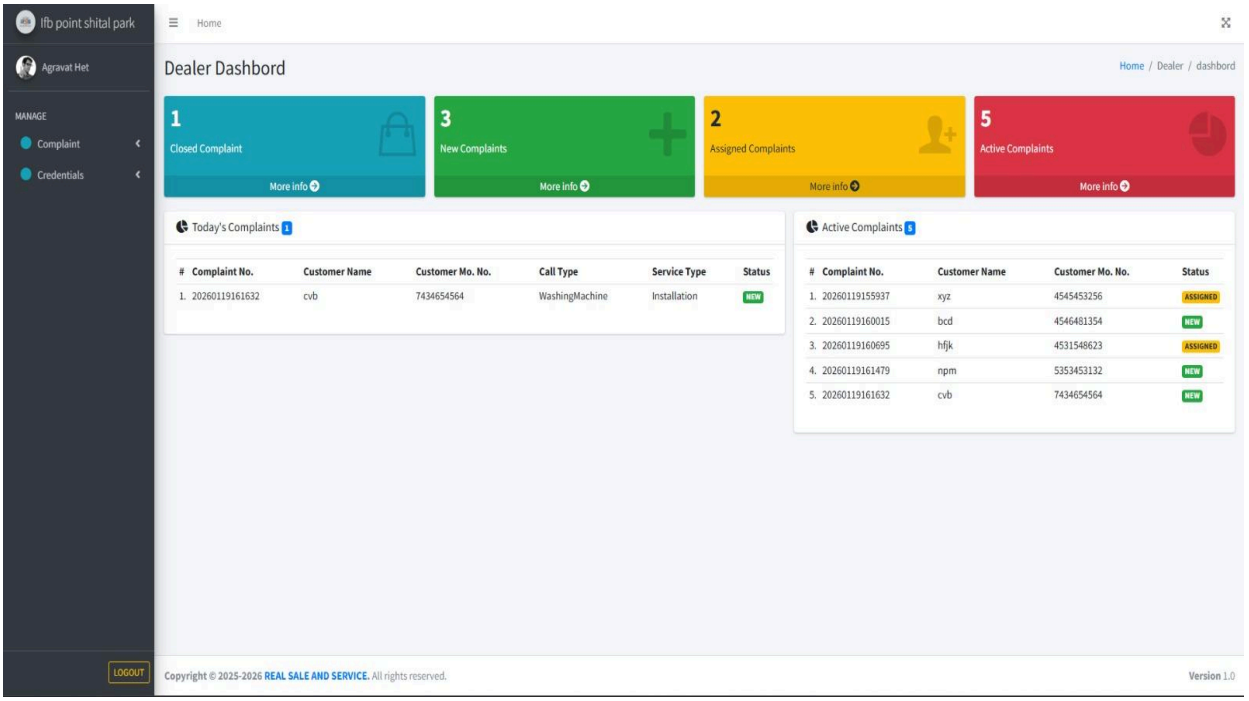



## 2. Dealer Dashboard

- **Description**: The Dealer Dashboard provides a quick overview of the dealer's activities within the SCMS. It highlights essential metrics such as the total number of complaints registered by the dealer, pending complaints, and recently updated complaints. The dashboard allows the dealer to efficiently manage service requests by tracking their status.
- **Operations**:

- **Overview:** Displays key statistics such as the number of complaints (new, in-progress, resolved).
- **Recent Complaints:** Lists recently registered or updated complaints for quick access.
- **Navigation:** Provides easy access to complaint management and dealer credentials sections.

- **Screenshot**:

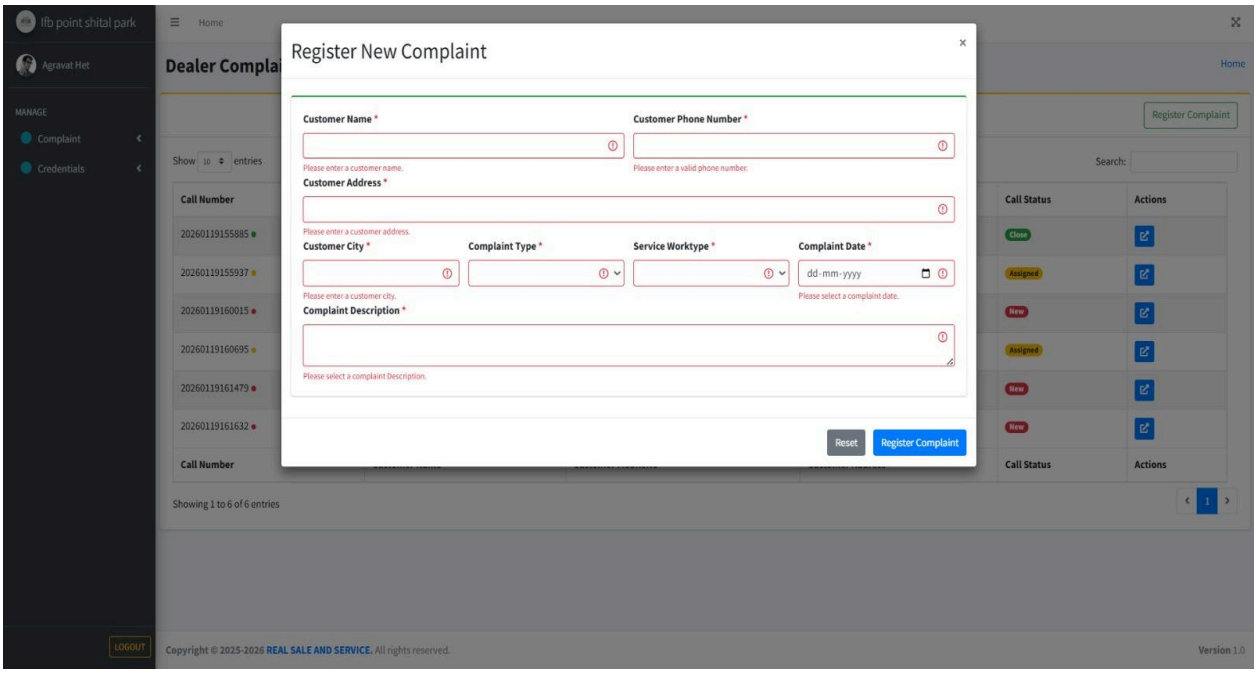


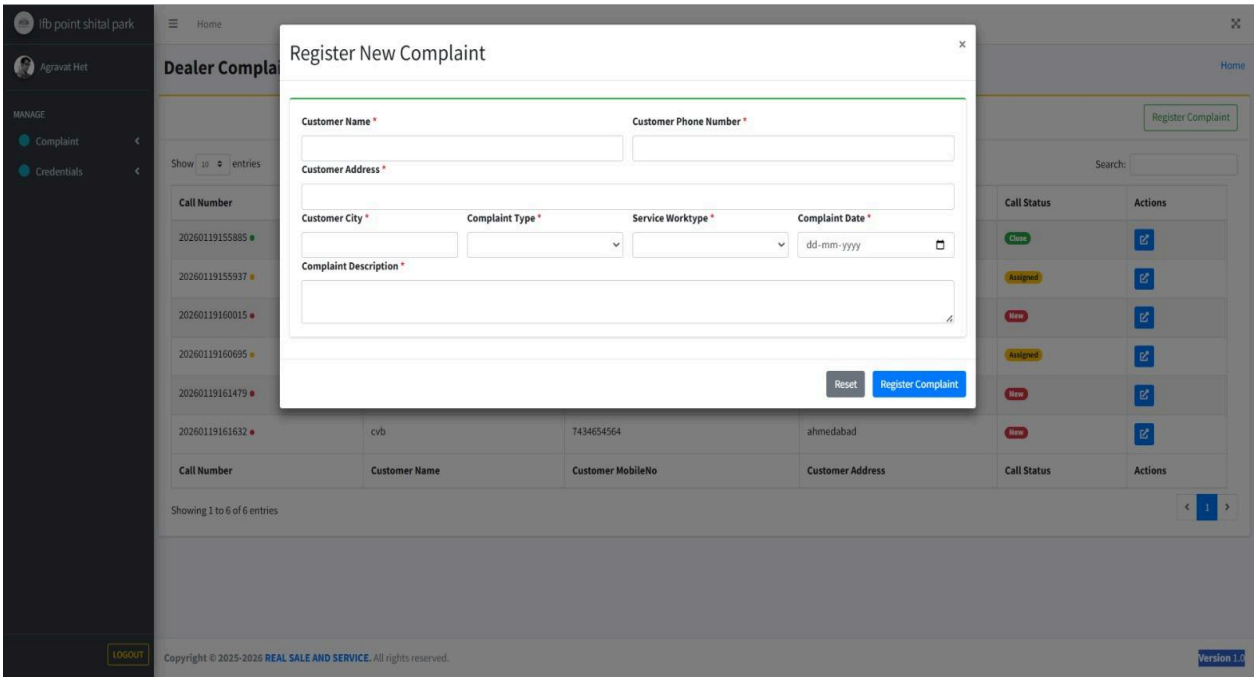

## 3. Complaint Management

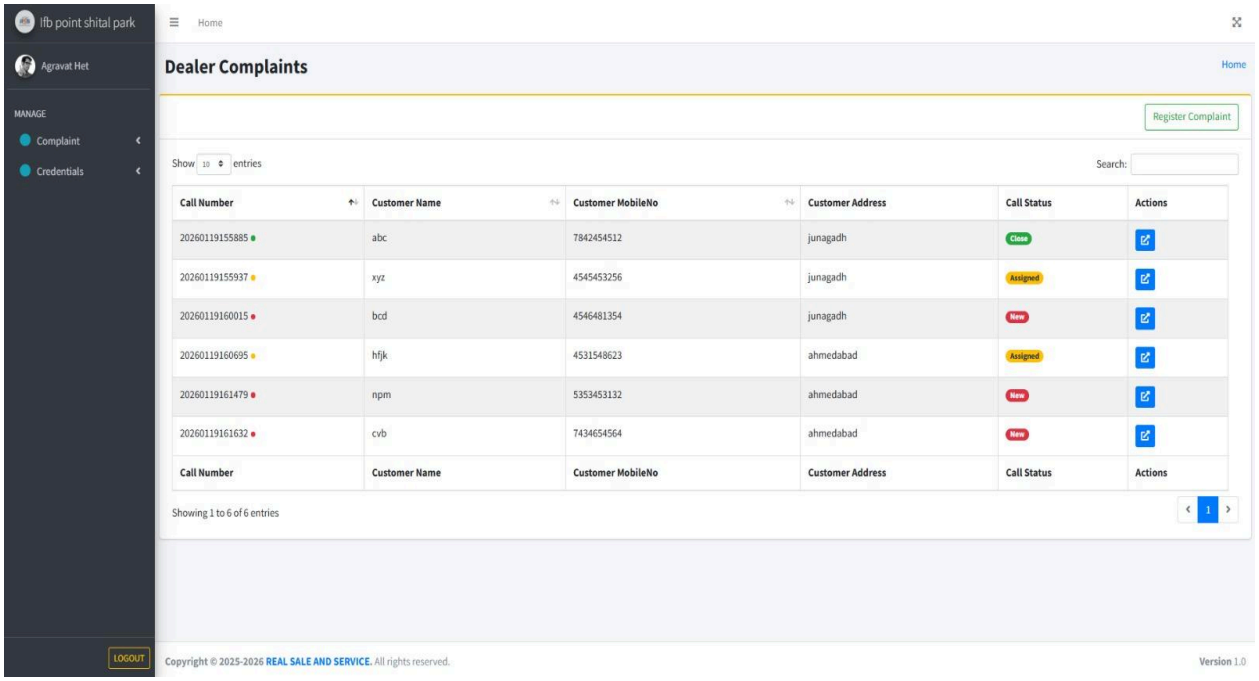
- **Description**: The Complaint Management module for Dealers allows them to register and track complaints on behalf of customers. Dealers can also view the status of existing complaints, ensuring they remain updated on the service progress and technician assignment.
- **Operations**:
  - **Insert:** Register new complaints by entering details like the customer's name, appliance type, and issue description.
  - **Show:** A table displays all registered complaints along with their current status (e.g., new, assigned, in-progress, closed).

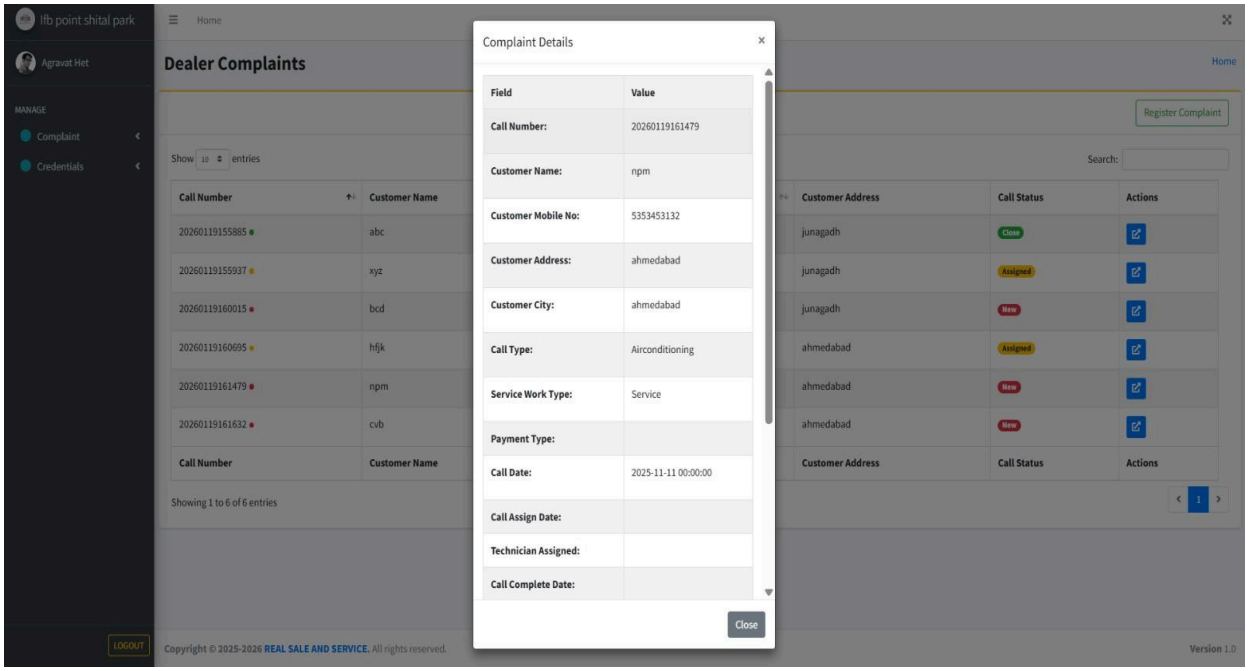
- **Update:** Dealers can update the complaint details (like issue description) before the complaint is processed, but once assigned, the status is handled by the Admin.

- **Screenshot**:



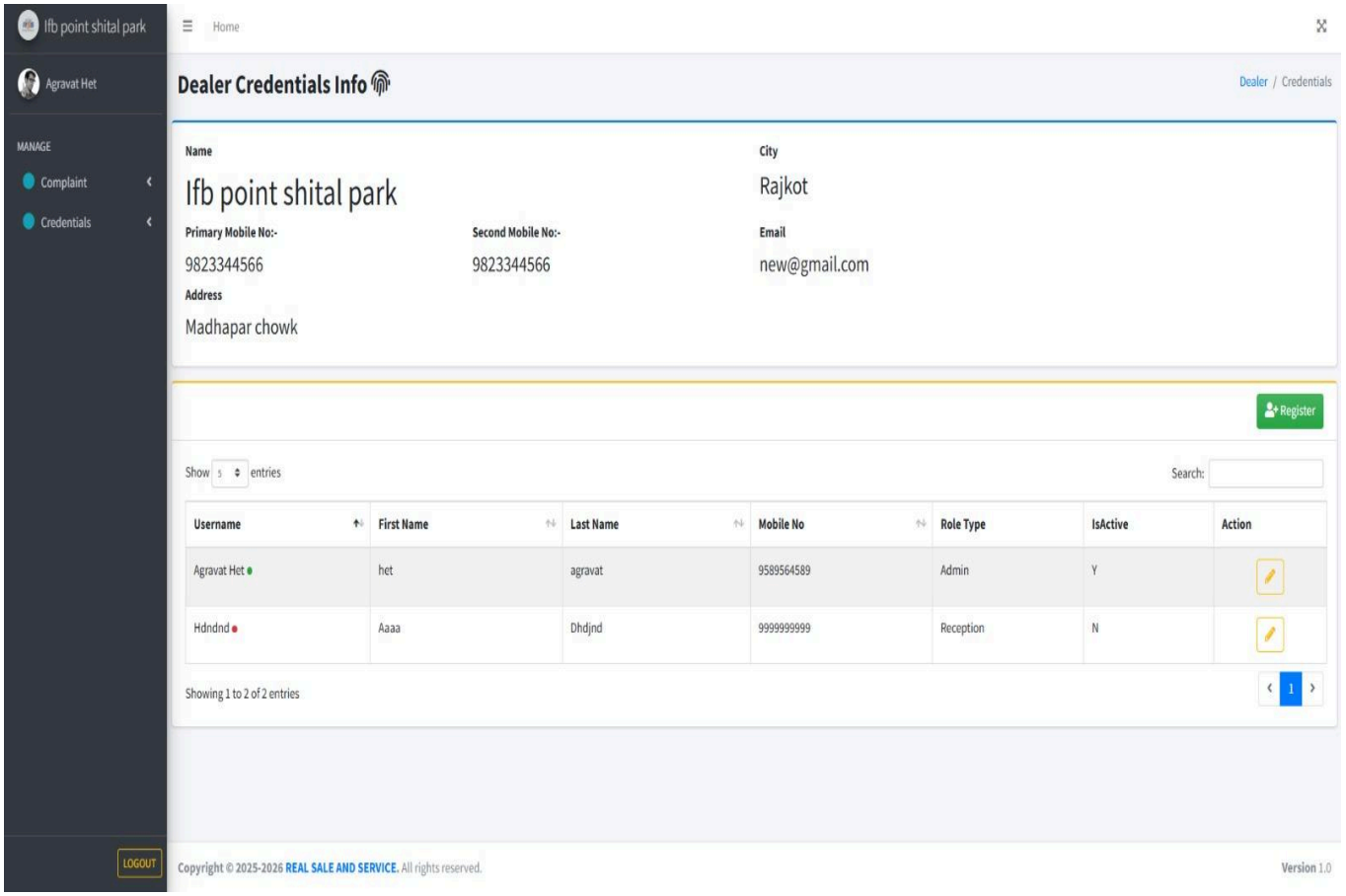

Ifb point shital park
Agravat Het
MANAGE
Complaint
Credentials
LOGOUT
Home
Dealer Complaints
Home
Register Complaint
Show 10 entries
Search:
Call Number
Customer Name
Customer MobileNo
Customer Address
Call Status
Actions
20260119155885 abc 7842454512 junagadh Close
20260119155937 xyz 4545453256 junagadh Assigned
20260119160015 bcd 4546481354 junagadh New
20260119160695 hfjk 4531548623 ahmedabad Assigned
20260119161479 npm 5353453132 ahmedabad New
20260119161632 cvb 7434654564 ahmedabad New
Showing 1 to 6 of 6 entries
1
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Version 1.0

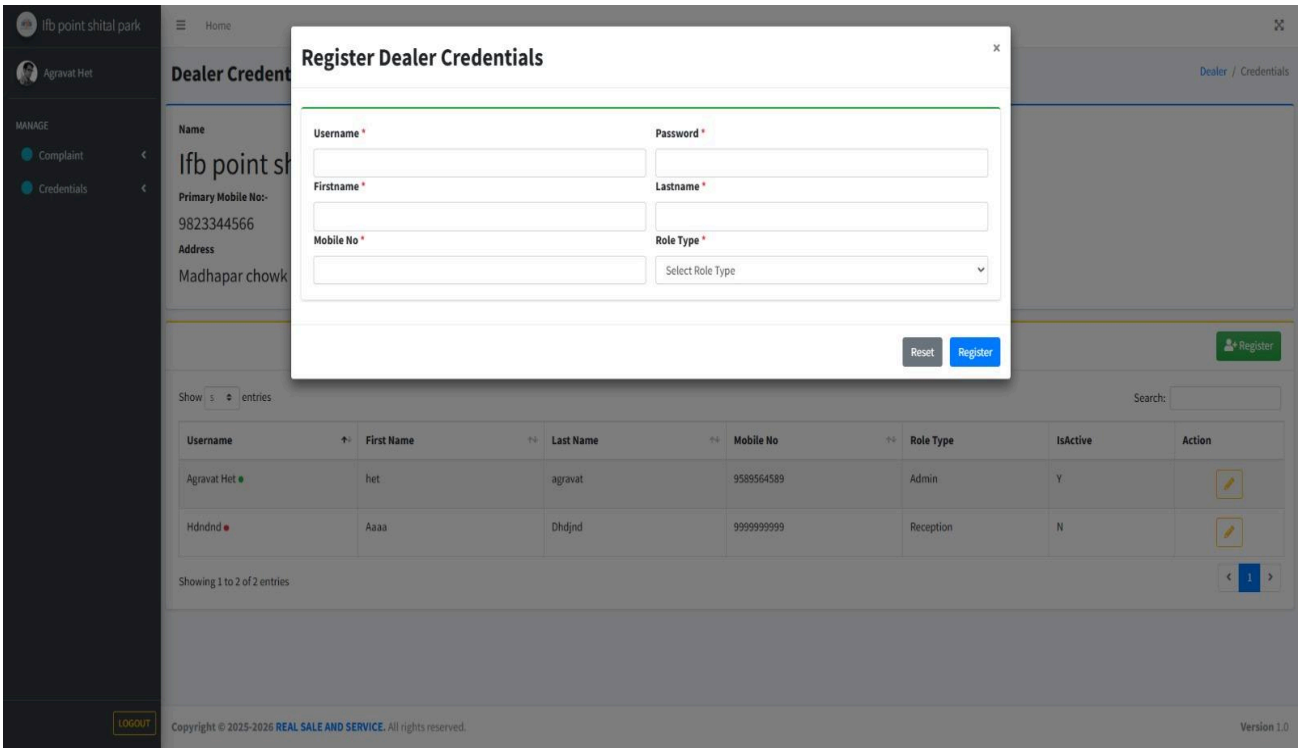

Complaint Details
Field
Value
Call Number: 20260119161479
Customer Name: npm
Customer Mobile No: 5353453132
Customer Address: ahmedabad
Customer City: ahmedabad
Call Type: Airconditioning
Service Work Type: Service
Payment Type:
Call Date: 2025-11-11 00:00:00
Call Assign Date:
Technician Assigned:
Call Complete Date:
Close

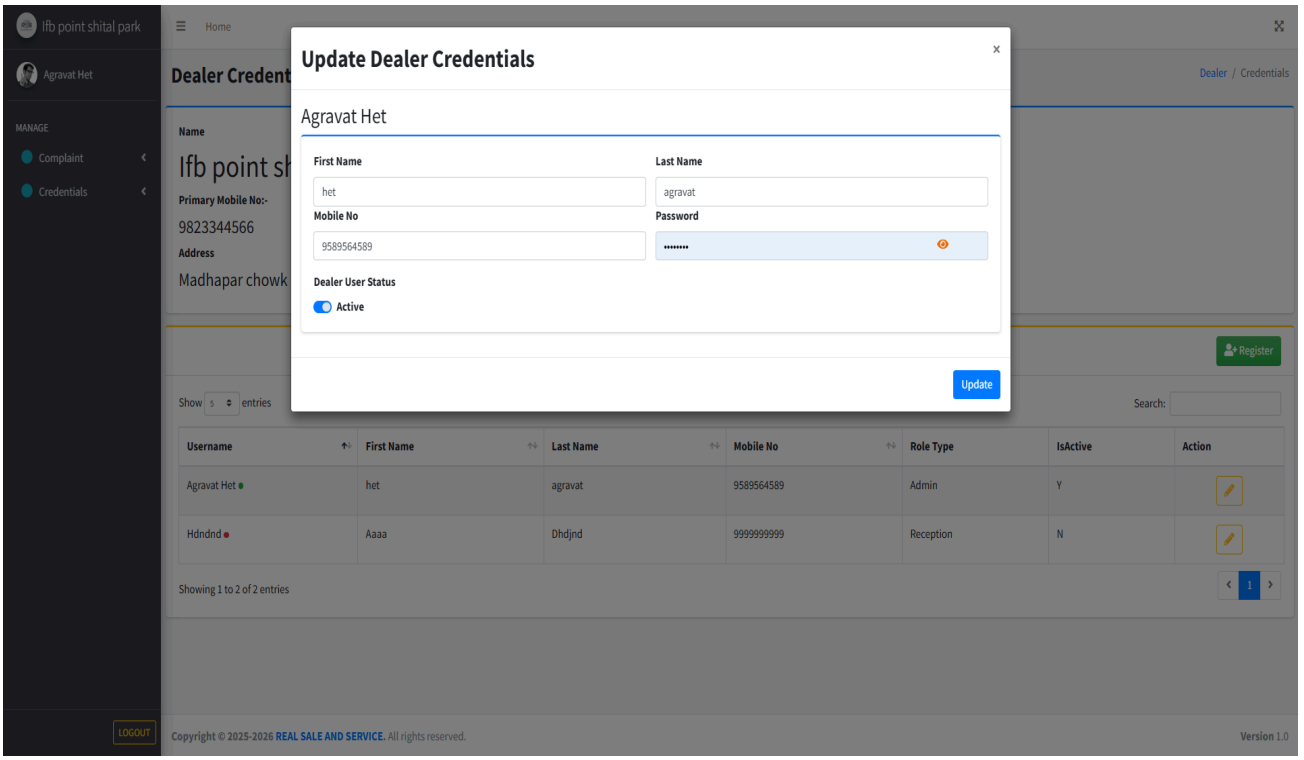
## 4. Dealer Credentials Management

- **Description**: Dealer Credentials Management enables dealers to manage their login information within the system. This section allows dealers to update their login credentials (e.g., change password) to ensure secure access to the SCMS.
- **Operations**:
    - **Show:** Displays the dealer's current login details (e.g., username) and last login time.
    - **Update:** Allows the dealer to change their password for better security, ensuring their account remains protected.

- **Screenshot**:




Register Dealer Credentials
Username *
Password *
Firstname *
Lastname *
Mobile No *
Role Type *
Select Role Type
Reset
Register


Update Dealer Credentials
Agravat Het
First Name
Last Name
het
agravat
Mobile No
Password
9589564589
Dealer User Status
Active
Update

# Testing

## Testing Standards:

Black box testing should make use of randomly generated inputs (only a test range should be specified by the tester), to eliminate any guess work by the tester as to the method of function.

Data outside of the specified input range should be tested to check robustness of the program.

Boundary cases should be tested (top and bottom of specified range) to make sure the highest and lowest allowable inputs produce proper output.

## Testing Strategies:

An efficient software testing or QA strategy requires testing of all technology stack levels to ensure that every, as well as the entire system, work without breaking down.

### 1. Unit Testing:

In computer programming, unit testing is a software verification and validation method in which a programmer tests if individual units of source code are fit for use. A unit test is the smallest test table part of an application. It is also called module testing.

### 2. Integration Testing:

In integration testing individual software modules are combined and tested as a group. It occurs after unit testing and before system testing. All the different modules of the project are combined and tested.

### 3. Module Testing:

Module testing is defined as a software testing type, which checks individual subprograms, subroutines, classes, or procedures of program.

- **Black Box Testing:**

- Black box testing is a software testing method where the internal structure, design, and implementation of the application under test are not known to the tester. Instead, the tester focuses on the functionality and behavior of the software. When it comes to testing Instant using black box testing, here are some approaches and considerations.

- **White Box Testing:**
  - White box testing, also known as structural testing or code-based testing, involves testing the internal structure, design, and implementation of a software application. In the case of Instant, white box testing would involve examining the source code and internal architecture of the application to identify potential issues and ensure that the code functions correctly. Here's how Instant could be tested through white box testing.

- **Gray Box Testing:**
  - Gray box testing is a testing technique that combines elements of both black box and white box testing. In gray box testing, testers have partial knowledge of the internal workings of the application under test, typically more than black box testing but less than white box testing. This approach allows testers to design test cases based on an understanding of the internal structure and logic of the application, while still focusing on the application's external behavior and functionality.

# Limitations and Future Enhancements

## Limitations :

While the Service Center Management System (SCMS) has successfully addressed several operational inefficiencies, it also has certain limitations:

1. **Limited User Access**:
    - Currently, only Admins and Dealers have access to the system. There is no direct interface available for technicians or customers, which restricts their ability to interact with the system.
2. **Basic Reporting**:
    - The existing system provides only basic reporting features. It lacks advanced analytics and detailed reporting capabilities that could offer deeper insights into service performance and efficiency.
3. **User Interface Constraints**:
    - The user interface is functional but may lack some advanced features and customizations that enhance user experience.
4. **Dependency on Internet Connectivity**:
    - The system requires a stable internet connection for optimal functionality, which may be a limitation in areas with poor connectivity.

## Future Enhancements :

To further improve the SCMS and expand its capabilities, the following enhancements are proposed:

1. **Technician Interface**:
    - Developing a separate interface for technicians to directly view and update their assigned complaints. This will enable them to manage their tasks more efficiently and keep the system updated in real time.
2. **Customer Portal**:
    - Implementing a customer portal that allows customers to directly register complaints and view the status of their service requests. This will enhance customer engagement and satisfaction.

3. **Mobile App**:

- Creating a mobile application for both Dealers and Admins to manage complaints on the go. A mobile app will provide flexibility and ease of access to the system.

4. **Advanced Reporting**:
   - Adding more detailed reporting features with visualizations such as graphs and charts to provide insights into technicians' performance, complaint resolution times, and overall service efficiency.

5. **Enhanced Security Measures**:
   - Implementing additional security features to protect user data and prevent unauthorized access to the system.

6. **Integration with Third-Party Services**:
   - Exploring the possibility of integrating the SCMS with third-party services for notifications (such as SMS and email alerts), further enhancing communication with users.

By addressing these limitations and implementing future enhancements, the SCMS can become a more comprehensive tool for managing service center operations, ultimately leading to improved customer satisfaction and operational efficiency.

# Bibliography

The following references were utilized to guide the development and implementation of the Service Center Management System (SCMS):


1. **Online Documentation**:
    - Bootstrap Documentation. Bootstrap 4 Getting Started.
      The official documentation for Bootstrap helped in understanding responsive design principles and implementing user-friendly interfaces.
    - jQuery Documentation. jQuery API Documentation.
      This documentation served as a reference for jQuery functions used for enhancing client-side interactions.
    - XAMPP Documentation. Apache Friends - XAMPP.
      This resource provided guidance on setting up a local development environment using XAMPP for PHP and MySQL projects.

2. **Additional Resources**:
    - Various online tutorials and forums for specific coding challenges and best practices related to PHP, MySQL, and JavaScript development.


These resources have played a crucial role in ensuring the successful development and deployment of the SCMS, equipping the team with the necessary tools and knowledge to address challenges effectively.